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**Retail Demand Forecasting Under Demand Heterogeneity: A
Demand-Type-Specific Study Using the Walmart M5 Dataset**

A project report submitted by: Rishab Kothari (6949800)

A project supervised by: Dr Dany Varghese

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University of Surrey
School of Computer Science and Electronic Engineering
Guildford, Surrey GU2 7XH
United Kingdom.
Tel: +44 (0)1483 300800

Abstract

Retail demand forecasting is difficult because products differ in how frequently they sell and how variable their sales quantities are. Forecasting results reported for a complete retail dataset may therefore hide substantial differences between products with smooth, erratic, intermittent and lumpy demand. This dissertation investigated how forecasting performance, feature contribution and fitted-model reliance varied across these four demand types using the Walmart M5 dataset.

All 30,490 item-store series were classified using Average Demand Interval and squared Coefficient of Variation following the framework of Syntetos, Boylan and Croston (2005). The classification identified 983 smooth, 494 erratic, 23,075 intermittent and 5,938 lumpy series. Intermittent and lumpy demand together represented 95.156% of the population. A balanced experimental sample containing 200 series from each demand type was used for model development and comparison.

Four forecasting models were evaluated: a seasonal naive baseline, Croston's method, the Syntetos–Boylan Approximation (SBA), and demand-type-specific XGBoost. All were assessed using five expanding walk-forward folds, each with a fixed 28-day test horizon. Weighted Root Mean Squared Scaled Error was used as the primary evaluation metric, with Mean Absolute Error and Root Mean Squared Error reported as secondary measures.

XGBoost produced the lowest mean WRMSSE for smooth demand (0.6360) and erratic demand (0.7938). Croston's method produced the lowest for lumpy demand (1.0674), while SBA produced the lowest for intermittent demand (1.0978). The demand-specific estimators outperformed the seasonal naive baseline wherever demand deviated from smooth regularity—via sparsity, variability, or both—not only for sparse demand. Croston's method reduced mean WRMSSE below the seasonal naive for erratic demand (0.8819 vs. 0.9500) despite erratic demand being frequent rather than sparse.

Feature contribution was investigated through leave-one-group-out ablation and exact TreeSHAP values across five feature groups. Calendar information provided the strongest ablation benefit for smooth and erratic demand, while price ranked first for intermittent and lumpy demand. Rolling features accounted for between 66.267% and 72.190% of absolute SHAP attribution but did not always correspond to forecasting benefit. The comparison between SHAP model reliance and ablation evidence is a core methodological contribution.

A restricted sequence-only LSTM feasibility experiment for smooth and erratic demand produced mean WRMSSE of 0.9278 and 1.2513 respectively—worse than both XGBoost and the seasonal naive baseline. The study concludes that retail forecasting models and feature sets should be evaluated separately by demand type, and that model reliance should be distinguished from out-of-sample predictive contribution when interpreting machine-learning forecasts. A classification stability audit confirmed that 93.8% of the 800 selected series retained their demand-type label when classification was restricted to training-only history, supporting the robustness of the demand-type-specific conclusions.

Keywords: retail demand forecasting, demand heterogeneity, Walmart M5, XGBoost, Croston, SBA, WRMSSE, feature ablation, SHAP, LSTM

Highlights

- XGBoost improved WRMSSE most for smooth and erratic demand.
- Croston/SBA outperformed XGBoost for intermittent and lumpy demand.
- Croston beat seasonal naive for erratic (frequent) demand too.
- Calendar features added most value for smooth and erratic demand.
- Price features were most useful for intermittent and lumpy demand.
- SHAP reliance and ablation benefit produced different rankings.

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Declaration

I certify that the work presented in the dissertation is my own unless referenced.

I confirm that the submitted work is my own work and that I have clearly identified and fully acknowledged all material that is entitled to be attributed to others (whether published or unpublished) using the referencing system set out in the programme handbook. I agree that the University may submit my work to means of checking this, such as the plagiarism detection service Turnitin UK. I confirm that I understand that assessed work that has been shown to have been plagiarised will be penalised.



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Chapter 1: Introduction

1.1 Background

Daily retail demand is not one forecasting problem. A product that sells five to ten units on most days presents a different challenge from one that sells nothing for a week and then records a single large order. The Walmart M5 dataset makes this visible at scale: of its 30,490 item-store series, 95.156% are classified as intermittent or lumpy under the ADI/CV² framework of Syntetos, Boylan and Croston (2005), and 68.0% of the 59,181,090 individual item-store-day observations record zero sales. A forecasting result reported only as a single aggregate score over the complete dataset would therefore be dominated by the behaviour of zero-heavy series and could conceal markedly different model performance across demand structures.

The fifth Makridakis forecasting competition confirmed that gradient-boosted tree models could produce substantial improvements over strong benchmarks in hierarchical retail forecasting, with the top five submissions improving Weighted Root Mean Squared Scaled Error by more than 20%. At the same time, approximately 92.5% of competing teams failed to outperform the strongest benchmark, and the improvements achieved by the leading methods were considerably smaller at the most disaggregated product-store levels (Makridakis, Spiliotis and Assimakopoulos, 2022). These aggregate findings leave an open question: do the gains from machine-learning models extend equally to smooth, erratic, intermittent and lumpy demand, or is the improvement concentrated in particular demand types?

A related gap concerns the interpretation of forecasting features. Retail forecasting models typically draw on recent sales history, calendar timing, event indicators and selling prices. Feature-importance measures such as SHAP (Lundberg and Lee, 2017) are often used to describe which predictors the fitted model relied upon. However, model reliance and out-of-sample predictive contribution are not the same property. A feature may receive a high SHAP attribution while providing no forecasting benefit after removal, or a low attribution while protecting the model against errors that matter for WRMSSE. Existing studies have not established whether this distinction differs across demand types.

This dissertation addresses these gaps by combining ADI/CV² demand classification, four-model forecasting comparison, within-type feature ablation, SHAP analysis and a limited LSTM feasibility experiment within one controlled design on the Walmart M5 dataset.

1.2 Research Aim and Objectives

The aim of this dissertation is to investigate how demand pattern type affects forecasting performance, feature contribution and model reliance in retail demand forecasting, using the Walmart M5 dataset as the empirical test bed.

The following five objectives operationalise this aim:

1. To classify all 30,490 item-store series in the Walmart M5 dataset into smooth, erratic, intermittent and lumpy demand types using the ADI/CV² framework of Syntetos, Boylan and Croston (2005), and to construct a balanced experimental sample of 200 series per demand type for controlled comparison.
2. To implement and evaluate four forecasting models — a seasonal naive baseline, Croston's method, the Syntetos–Boylan Approximation (SBA) and a demand-type-specific XGBoost model — across five expanding walk-forward

folds, each with a fixed 28-day test horizon, using Weighted Root Mean Squared Scaled Error as the primary metric.

3. To quantify the contribution of five feature groups to XGBoost forecast accuracy using a leave-one-group-out ablation design, and to compare ablation-based predictive contribution against TreeSHAP model reliance for each demand type.

4. To assess the feasibility of a sequence-only LSTM architecture as a neural forecasting baseline for smooth and erratic demand, and to compare its performance against the statistical and tree-based models evaluated in Objective 2.

5. To evaluate the robustness of the demand-type-specific findings through a classification stability audit, an alpha-sensitivity analysis for Croston and SBA, a WRMSSE denominator audit, and a within-sample cohort stability check.

These five objectives map directly to five research questions that structure the results and discussion chapters:

RQ1: How does demand type affect the relative forecasting accuracy of seasonal naive, Croston, SBA and XGBoost models?

RQ2: Which feature groups contribute most to XGBoost forecast accuracy for each demand type, as measured by leave-one-group-out ablation?

RQ3: How does TreeSHAP model reliance compare to ablation-based predictive contribution across demand types, and where do they diverge?

RQ4: Can a sequence-only LSTM architecture match the performance of XGBoost or statistical baselines for smooth and erratic demand?

RQ5: How can the combined model-performance, feature-ablation and SHAP results inform practical model-selection guidance for retail demand planning?

1.3 Research Approach

The study follows a quantitative comparative forecasting design using the Walmart M5 dataset. All 30,490 item-store series are first classified into the four demand types using ADI and CV². A balanced experimental subset of 200 series per type (800 total, 1,552,800 item-store-day observations) is then used for model development and evaluation. The balanced design is a deliberate experimental choice to make each demand type equally visible in the comparison; it is not representative sampling, and this constraint is stated as a limitation throughout.

Four forecasting models are compared: a seasonal naive benchmark that repeats the final seven training days across the 28-day horizon; Croston's method (Croston, 1972) and the Syntetos–Boylan Approximation (Syntetos and Boylan, 2005), both demand-specific estimators designed for irregular demand; and demand-type-specific XGBoost models trained with 18 predictors organised into lag, rolling, calendar, event/SNAP and price groups. All models are evaluated using five expanding walk-forward folds with a fixed 28-day test window. WRMSSE is the primary metric; MAE and RMSE are retained as secondary measures.

Feature contribution is assessed through leave-one-group-out ablation, in which one feature group at a time is removed and the XGBoost model is retrained under the locked design. Exact TreeSHAP values are calculated for all 112,000 final XGBoost forecasts and aggregated into the same five feature groups. The comparison between SHAP reliance and ablation benefit is treated as a core analytical contribution.

A restricted sequence-only LSTM feasibility experiment is conducted for smooth and erratic demand. The LSTM receives only the previous 56 daily sales observations and produces a direct 28-day forecast. It does not receive the calendar, event, SNAP or price information available to XGBoost. This constraint is stated every time the LSTM result is reported.

All comparisons are descriptive. Five forecast origins are insufficient for a valid Diebold–Mariano test, and individual series-day errors share forecast origins and models. Formal statistical significance is not claimed.

1.4 Dissertation Outline

Chapter 2 reviews the literature on retail forecasting, the M5 competition, demand classification, machine learning, feature engineering, SHAP, forecast evaluation and deep-learning forecasting. It identifies the research gap that motivates the demand-type-specific design.

Chapter 3 describes the research approach, including dataset preparation, the ADI/CV² classification procedure, balanced-subset construction, feature engineering, the specifications of all four forecasting models, the walk-forward validation design, WRMSSE calculation, ablation protocol, SHAP implementation and the LSTM feasibility design.

Chapter 4 presents the empirical results: demand classification, exploratory analysis, seasonal naive benchmark, Croston and SBA results, XGBoost performance, the four-model comparison, hierarchy-level evaluation, feature-ablation findings, SHAP model-reliance analysis, the SHAP-versus-ablation comparison and the LSTM feasibility outcome.

Chapter 5 discusses the findings in relation to the literature and research questions, addresses the Croston/SBA surprise for erratic demand, interprets the SHAP-versus-ablation divergence, and considers practical implications, limitations and interpretation boundaries.

Chapter 6 concludes the dissertation by answering each research question, summarising the contributions, outlining future research directions and providing personal reflections on the project.

Chapter 2: Literature Review

2.1 Retail Demand Forecasting Background

Retail demand forecasting supports decisions at strategic, tactical and operational levels. At the strategic level, forecasts inform decisions concerning store locations, sales channels, target markets and distribution networks. Tactical forecasts support decisions about product categories, assortment planning, pricing and promotional activity. At the operational level, short-term forecasts are required for store replenishment, purchasing, distribution, workforce planning and day-to-day inventory control. These levels are connected because decisions made at the product and store level eventually affect wider supply-chain and business performance (Fildes, Ma and Kolassa, 2022).

Forecasting errors can create direct operational problems for retailers. Underforecasting may result in stock shortages, reduced product availability, lost sales and lower customer satisfaction. Overforecasting may lead to unnecessary inventory, additional storage costs, waste and product write-offs. This is especially important in high-volume and low-margin retail environments, where relatively small forecasting errors can affect a large number of products and locations. Reliable forecasts are therefore needed not only to estimate expected sales but also to coordinate purchasing, distribution and replenishment activities across the retail network (Fildes, Ma and Kolassa, 2022).

The difficulty of retail forecasting increases as the level of detail becomes more granular. Sales aggregated across a company, state or product category may display recognisable trend and seasonal behaviour. In contrast, demand at the individual stock-keeping-unit and store level can contain many zero-sales observations, sudden spikes and irregular changes. Product-level demand may also be affected by stock-outs, weekly seasonality, holidays, calendar events, prices, promotions and differences between stores. These factors mean that a forecasting method that performs well for aggregated demand may not provide the same level of accuracy for individual item-store series. The level at which forecasts are produced must therefore match the decision that the forecast is intended to support (Fildes, Ma and Kolassa, 2022).

Retail forecasting can also involve a large number of possible explanatory variables. A product's sales may be affected by its own previous sales and price, but they may also respond to promotions applied to related products. Ma, Fildes and Huang (2016) examined weekly data covering 926 stock-keeping units across 15 food and drink categories over 320 weeks. Their study used sales history, price, displays, advertising, calendar events and information from related products. Under rolling evaluation, their proposed models improved forecast accuracy by approximately 12.6% compared with a model using only the focal product's predictors. Around 95% of this additional improvement came from information within the same product category, while only around 5% came from other categories. These results show that additional information can be useful, but its value depends on careful feature selection and repeated temporal evaluation. Simply adding more variables does not guarantee better forecasts and may increase the risk of overfitting (Ma, Fildes and Huang, 2016).

The Walmart M5 competition illustrates the scale and complexity of modern retail forecasting. The dataset contains sales for 3,049 products, producing 30,490 product-store series and 42,840 series across 12 hierarchical aggregation levels. Forecasts were evaluated over a 28-day horizon. The strongest machine-learning methods achieved improvements of more than 20% over the strongest competition benchmark. However, approximately 92.5% of participating teams failed to outperform that

benchmark. The reported improvements were also smaller at the most disaggregated product and product-store levels. This shows that machine-learning complexity does not automatically produce more accurate forecasts and that strong baseline methods remain necessary when evaluating a new forecasting approach (Makridakis, Spiliotis and Assimakopoulos, 2022).

The presence of zero and near-zero observations also affects how retail forecasts should be evaluated. Percentage-based measures such as the Mean Absolute Percentage Error can become undefined or misleading when actual sales are zero. Hyndman and Koehler (2006) therefore recommended the use of scaled errors when comparing forecasting performance across series with different sales levels. Scaled measures compare forecast errors with the errors produced by a simple benchmark, allowing results from differently scaled series to be interpreted more fairly. This supports the use of an established baseline before assessing whether a more complex model provides genuine forecasting improvement (Hyndman and Koehler, 2006).

The literature therefore presents retail forecasting as a problem shaped by data granularity, demand irregularity, explanatory-variable availability and the choice of evaluation method. Existing evidence supports the use of retail features and machine-learning models, but it also shows that their benefits are not consistent across all products or forecasting levels. This creates a need to evaluate forecasting models separately for different forms of demand rather than relying only on one aggregate performance score. The next section introduces the Walmart M5 dataset and explains how the competition provides the empirical setting for this dissertation.

2.2 Walmart M5 Dataset and Forecasting Competition

The Walmart M5 dataset was released for the fifth Makridakis forecasting competition and provides a large-scale example of hierarchical retail demand forecasting. Unlike earlier M competitions, which included time series from several industries and frequencies, the M5 competition focused specifically on daily retail sales. This made it possible to examine forecasting methods in a setting that closely reflects the operational problems faced by large retailers, including product-level demand variation, calendar effects, price changes and irregular sales patterns.

The dataset contains sales for 3,049 Walmart products sold across ten stores in three US states: California, Texas and Wisconsin. Combining each product with its selling location produces 30,490 product-store series at the lowest level. These series can be aggregated into 42,840 time series across 12 hierarchical levels, including total sales, state, store, category, department, product and product-store groupings. This structure allows forecasting performance to be evaluated at both highly aggregated and highly disaggregated levels (Makridakis, Spiliotis and Assimakopoulos, 2022).

The M5 data are organised into three main files. The sales file contains daily unit sales for each product-store combination. The calendar file connects each sales day to its date, weekday, month and year, as well as named events and Supplemental Nutrition Assistance Program indicators for the three states. The price file contains weekly selling prices for each item and store. Together, these sources allow models to use both historical sales and explanatory retail variables. This is important because retail demand may be influenced by weekly seasonality, public events, changes in price and differences between stores or states.

Participants were required to produce forecasts for a 28-day horizon. Forecasts were submitted for the 30,490 product-store series and then aggregated to the remaining hierarchy levels. This ensured that the forecasts were coherent, meaning that forecasts at the lower levels summed to the forecasts reported at higher levels. Forecasting

accuracy was measured using the Weighted Root Mean Squared Scaled Error. RMSSE scales the squared forecast error for each series using the in-sample error of a one-step naive forecast. This scaling allows series with different sales volumes to be compared more fairly. The final WRMSSE score combines the RMSSE values across all 42,840 series using weights based on the dollar sales generated during the final 28 days of the training period. Products and aggregated groups associated with greater recent revenue therefore receive greater influence in the final score (Makridakis, Spiliotis and Assimakopoulos, 2022).

The competition results demonstrated the potential of machine-learning methods in retail forecasting. The top 50 submissions were based on machine-learning approaches, and the five winning teams improved WRMSSE by more than 20% compared with the strongest benchmark. Gradient-boosted decision-tree models, particularly LightGBM, were widely used by the leading teams. The winning method combined several LightGBM models trained using different groupings and forecasting strategies.

Several broader lessons emerged from the competition. Leading teams commonly used cross-learning, where one model learns from multiple related series. Combinations of multiple models generally performed better than relying on one model. The strongest teams used several rolling 28-day validation windows rather than selecting their models from one validation period. However, the competition did not show that machine learning always produces more accurate forecasts. Approximately 92.5% of the participating teams failed to outperform the strongest exponential-smoothing benchmark. Performance also differed considerably across the hierarchy: improvements over the benchmark were approximately 40% at the total level but only around 3% at the product, product-state and product-store levels (Makridakis, Spiliotis and Assimakopoulos, 2022).

This variation is directly relevant to the present dissertation. The M5 competition established that machine-learning methods can produce substantial overall improvements, but its aggregate results do not fully explain how those improvements vary between different demand patterns. This dissertation therefore uses the M5 data to investigate whether forecasting performance, feature contribution and model reliance differ across smooth, erratic, intermittent and lumpy demand.

Makridakis, Spiliotis and Assimakopoulos (2020) report a similar pattern in the M4 competition, which preceded M5 as the benchmark forecasting study for the wider series. Comparing 61 methods across 100,000 series, they found that purely machine learning methods performed poorly relative to statistical and hybrid approaches, and that the overall winning method combined an exponential smoothing component with a recurrent neural network rather than relying on either technique alone. The gap was wide enough that machine learning methods did not decisively outperform statistical methods until the subsequent M5 competition. The present study's finding that no single model (naive, Croston, SBA, or XGBoost) dominates across all demand types echoes this broader pattern: in both M4 and this study, forecasting accuracy depends more on matching method to the structure of the underlying series than on any general superiority of machine learning over statistical approaches.

2.3 Demand Heterogeneity and ADI/CV² Classification

The ADI/CV² framework proposed by Syntetos, Boylan and Croston (2005) remains the most widely used demand classification scheme in the intermittent-demand forecasting literature. Its two dimensions — demand frequency measured by Average Demand Interval and demand-size variability measured by squared Coefficient of Variation — produce four categories that have been shown to predict relative method

performance in controlled experiments. However, several limitations of the framework are relevant to the present study and must be acknowledged before its application is justified.

The original cut-off values of $ADI = 1.32$ and $CV^2 = 0.49$ were derived empirically from 3,000 monthly automotive spare-parts series, each containing only 24 observations. The transfer of these thresholds to daily grocery retail data — where series contain up to 1,941 observations, demand intervals are measured in days rather than months, and product lifecycles differ substantially from industrial spare parts — is not theoretically guaranteed. Fildes, Ma and Kolassa (2022) note that intermittence is a common feature of daily SKU-level point-of-sale data, but they do not confirm that the Syntetos-Boylan-Croston thresholds produce optimal category boundaries in that setting. The present study applies the framework operationally, using the established cut-offs as a classification instrument rather than treating the category boundaries as natural demand-structure thresholds.

A second limitation concerns classification stability over time (Januschowski et al., 2020). A series classified as intermittent using its complete sales history may have been erratic or lumpy at an earlier forecast origin, when less history was available. This temporal instability is particularly acute for series near the $ADI = 1.32$ or $CV^2 = 0.49$ boundaries. In the present study, a classification stability audit found that 93.8% of the 800 selected series retained their full-history demand-type label when classification was restricted to training observations available before the earliest forecast origin. Agreement was highest for intermittent demand (98.5%) and lowest for erratic demand (89.5%). Of the 800 series, 415 fell within 0.10 of at least one threshold, confirming that a substantial minority of series sit in ambiguous regions where small changes in available history can alter the assigned class. These borderline series represent a genuine source of uncertainty in the demand-type-specific conclusions, particularly for erratic demand.

A third limitation is that the framework uses fixed global thresholds rather than thresholds estimated from the data at hand. Alternative classification approaches — including continuous ADI/CV^2 characterisation without hard boundaries, time-varying classification at each forecast origin, and occurrence-size decomposition models that treat demand arrival and demand size as separate processes — have been proposed in the literature but were not adopted here. The fixed-threshold design was retained for comparability with the established literature and because the stability audit indicated that classifications were stable for the demand types carrying the study's strongest conclusions (98.5% agreement for intermittent, 92.5% for lumpy).

Table 2.1 summarises the four demand categories produced by the framework as applied in this study.

Table 2.1: ADI/ CV^2 demand classification framework

Demand type	ADI condition	CV^2 condition	Interpretation
Smooth	$ADI \leq 1.32$	$CV^2 \leq 0.49$	Frequent demand, lower variation
Erratic	$ADI \leq 1.32$	$CV^2 > 0.49$	Frequent demand, higher variation
Intermittent	$ADI > 1.32$	$CV^2 \leq 0.49$	Infrequent demand, stable quantities
Lumpy	$ADI > 1.32$	$CV^2 > 0.49$	Infrequent demand, variable quantities

2.4 Croston's Method and the Syntetos–Boylan Approximation

Croston (1972) proposed a forecasting method specifically designed for intermittent demand. The method separates the forecasting problem into two components: the size of non-zero demand and the interval between non-zero demand occurrences. Each component is estimated separately using exponential smoothing, and the forecast is obtained by dividing the estimated demand size by the estimated inter-demand interval. This decomposition allows the method to avoid forecasting zero explicitly while still producing a mean demand estimate that reflects both the frequency and magnitude of positive sales.

Syntetos and Boylan (2001) identified a positive bias in Croston's original estimator and proposed the Syntetos–Boylan Approximation (SBA) as a correction. SBA applies a multiplicative adjustment factor to the Croston forecast that reduces the upward bias present when the smoothing parameter is non-zero and the inter-demand interval is estimated. The adjustment is derived from the mathematical expectation of the ratio estimator used by Croston's method. Both methods share the same underlying decomposition; SBA modifies only the final recombination step. Recent evaluations confirm that SBA remains competitive against more complex intermittent-demand methods on daily retail data, though performance gains over simpler benchmarks are often modest and context-dependent (Petropoulos et al., 2022).

These methods were originally evaluated on monthly industrial-parts data, where demand was genuinely sparse and the ADI was typically well above the 1.32 threshold. Neither method was designed with frequent daily retail demand in mind. Their inclusion in this dissertation allows the study to test whether demand-specific estimators designed for irregular demand provide forecasting value in a daily retail context—not only for the intermittent and lumpy series where they were originally intended, but also for the frequent-demand categories where their behaviour has received less empirical attention (Svetunkov and Boylan, 2023).

2.5 Machine Learning and XGBoost in Demand Forecasting

Machine-learning forecasting normally converts the time-series problem into a supervised-learning structure. Each observation is represented by a target value and a set of predictors derived from information available before the forecast origin. Tree-based models are suitable for structured retail data because they divide the predictor space into a sequence of decision regions. Gradient boosting addresses the instability of single decision trees by building trees sequentially, with each new tree trained to reduce errors remaining from the trees already included in the model.

Chen and Guestrin (2016) introduced XGBoost as a scalable gradient-tree-boosting system. The learning objective combines the forecasting loss with a penalty for model complexity, and the framework uses second-order optimisation, shrinkage, column subsampling and sparsity-aware tree construction. XGBoost was designed for large tabular datasets, but the original evaluation used classification, ranking and prediction tasks rather than a retail time-series experiment.

XGBoost has several characteristics that make it relevant to retail demand data. It can model nonlinear effects without requiring each relationship to be specified in advance, process numerical, binary and encoded categorical inputs within one framework, and handle missing values natively within its tree-building process. The M5 competition provided stronger evidence for gradient-boosting methods in large-scale retail forecasting, with leading submissions commonly using LightGBM (Makridakis, Spiliotis and Assimakopoulos, 2022).

Retail forecasting research does not support the assumption that complex methods will always outperform simple ones. Fildes, Ma and Kolassa (2022) found that multivariate and machine-learning methods had shown promise, but the available evaluations differed in datasets, forecasting horizons, product groups and benchmarks. The form of the target distribution is another consideration. Retail sales are non-negative, often contain many zeros and may be strongly right-skewed. Zhou, Qian and Yang (2022) examined gradient-tree boosting with a Tweedie distribution for zero-heavy and non-negative outcomes. Their specialised EMTboost model is not the same as the standard Tweedie objective available in XGBoost, but the paper supports the broad distributional reasoning behind a Tweedie objective.

XGBoost was selected as the primary machine-learning model in this study because the M5 data can be represented as structured observations using sales-history and known retail features. The model offers a manageable balance between nonlinear predictive capacity, computational feasibility and compatibility with feature-based interpretation.

2.6 Feature Engineering and Ablation in Forecasting

Machine-learning models require retail time-series data to be converted into a supervised-learning format. Historical sales features, including lag variables and rolling summaries, capture patterns already present in the demand series. The usefulness of these features may depend on the demand pattern: smooth demand provides frequent non-zero observations, while intermittent demand contains longer zero-sales intervals. Calendar features represent repeating time structures, event variables provide information about named occasions, and price is another common retail predictor (Fildes, Ma and Kolassa, 2022).

Ma, Fildes and Huang (2016) provide a detailed example of forecasting with high-dimensional retail features. Under their rolling forecasting scheme, models using additional information improved accuracy by approximately 12.6% compared with a model using only the focal SKU's predictors. Around 95% of the additional improvement came from products in the same category. Their study also showed that additional information did not always improve forecasting, and high-dimensional predictor sets can contain redundant and noisy variables.

Feature importance from a fitted model cannot resolve the question of predictive value by itself. A model may frequently rely on one feature because it is available for splitting or because it is correlated with other predictors. A feature-ablation study provides a direct way to examine forecast benefit: the full model is first trained using all feature groups, one group is then removed while the remaining modelling procedure is held constant, and the change in out-of-sample error shows whether performance improved or worsened. The present dissertation applies ablation separately to lag, rolling, calendar, event/SNAP and price groups within each demand type.

2.7 Explainability: SHAP and Tree-Based Forecasting Models

Lundberg and Lee (2017) introduced SHAP as a unified framework for explaining model predictions based on Shapley values from cooperative game theory. Each feature receives a contribution showing how it moves the prediction away from a baseline expected value. The framework satisfies three properties — local accuracy, missingness and consistency — that earlier importance measures such as permutation importance and split-count heuristics do not jointly satisfy. Lundberg et al. (2020) extended this framework for tree-based models through TreeExplainer, which calculates exact Shapley values in polynomial time using the internal tree structure rather than sampling-based approximations.

Despite these theoretical properties, several interpretation boundaries are critical for the present study. The most important distinction is between model reliance and predictive contribution. A high mean absolute SHAP value indicates that the fitted model allocates substantial prediction credit to a feature — it does not indicate that retaining the feature improves out-of-sample forecast accuracy. These two quantities answer different questions. SHAP attribution is a property of the fitted model on the training and validation data; ablation-based predictive contribution is a property of the feature's marginal out-of-sample value under retraining. Kumar et al. (2020) demonstrate formally that conditional Shapley methods, including TreeSHAP, estimate feature importance from the model's behaviour across training examples where that feature takes a particular value, rather than from a direct interventional test of what happens when the feature is removed. When predictors are correlated, the feature that is structurally easier for the tree to split on can absorb credit that belongs to a correlated predictor with greater genuine predictive value.

This distinction is directly relevant to retail forecasting with lag and rolling features. Lag and rolling windows are highly correlated with one another and with the target variable for smooth demand. For sparse demand, these same features mostly encode strings of zeros, yet the tree may still split on them because they carry structural information about whether a series is currently active. A high SHAP share for rolling features in an intermittent-demand model may therefore reflect the model's use of recent-history windows as a proxy for demand-occurrence state rather than evidence that rolling statistics improve forecasting accuracy. The leave-one-group-out ablation design used in this study provides a direct empirical test of whether removing each feature group improves or worsens out-of-sample WRMSSE, which is a different and complementary question to what SHAP attribution measures.

A further boundary concerns causal interpretation. SHAP values explain the fitted model rather than the underlying demand process. A feature receiving high SHAP importance did not necessarily cause the observed sales outcome; it may simply be correlated with the true driver or serve as a proxy in the model's learned representation. Claims about why demand behaves as it does cannot be supported by SHAP evidence alone and require either experimental manipulation or structural economic reasoning outside the scope of this study. The present dissertation treats SHAP values as a measure of fitted-model reliance and compares them explicitly with ablation-based forecast benefit, following the recommendation of Kumar et al. (2020) that attribution and predictive-contribution evidence should be evaluated separately.

2.8 Hierarchical Forecast Evaluation, WRMSSE and Walk-Forward Validation

Forecasting performance cannot be assessed reliably without an evaluation design that reflects how the model will be used after development. The M5 Accuracy competition evaluated 30,490 product-store series and their aggregation into 42,840 series across 12 hierarchical levels using WRMSSE. Hyndman and Koehler (2006) demonstrated that several widely used percentage and relative accuracy measures can become infinite, undefined or misleading in ordinary forecasting situations, and argued that scaled errors are more suitable for comparisons across multiple series.

Tashman (2000) argued that forecasting methods should be tested using observations excluded from model fitting and recommended rolling origins, model recalibration and multiple test periods. The present dissertation uses five expanding walk-forward folds with a 28-day test window in each fold, with results reported separately for smooth, erratic, intermittent and lumpy demand. WRMSSE is the primary measure; MAE and RMSE are retained as secondary measures.

2.9 LSTM and Deep Learning for Demand Forecasting

Hochreiter and Schmidhuber (1997) introduced the Long Short-Term Memory architecture to address the vanishing-gradient problem in recurrent neural networks. LSTM uses memory cells and gating mechanisms to control information flow across long sequences. The original evaluation used artificial sequence-learning tasks and did not involve retail demand or multi-step forecasting.

The application of recurrent neural networks to large-scale retail forecasting has advanced substantially since the basic LSTM architecture. Salinas et al. (2020) introduced DeepAR, a global autoregressive recurrent model that trains across a collection of related time series and produces probabilistic forecasts using appropriate likelihoods, feature scaling, weighted sampling and known future covariates (Wang et al., 2019). DeepAR demonstrated that global neural forecasting — where one model learns from many series simultaneously — can outperform local statistical models when sufficient related series are available (Bandara, Bergmeir and Smyl, 2020). Oreshkin et al. (2020) introduced N-BEATS, a pure deep learning architecture based on backward and forward residual links and interpretable basis expansions, which achieved strong performance on the M4 competition without using any time-series-specific components. These architectures differ fundamentally from a basic sequence-to-sequence LSTM in their use of cross-series learning, probabilistic objectives, covariate integration and architectural depth (Herzen et al., 2022).

The M5 competition results are informative about the conditions under which neural methods add value in retail forecasting. The top submissions used gradient-boosted tree models rather than neural networks as their primary forecasting engine, with neural components appearing mainly in ensemble stacking layers. This does not establish that neural forecasting is inferior in general; it reflects that the M5 evaluation favoured methods that could effectively use the structured tabular features — calendar, event, price and cross-series lag information — that tree-based models handle natively. Neural architectures that incorporate these features as known future covariates, such as DeepAR or Temporal Fusion Transformers (Lim et al., 2021), were not among the top M5 submissions but have shown competitive performance in subsequent retail forecasting evaluations.

The LSTM experiment in this dissertation is explicitly restricted and should not be interpreted as a general evaluation of neural forecasting for retail demand. The sequence-only architecture received only the previous 56 daily sales observations and produced a direct 28-day forecast. It did not receive calendar, event, SNAP or price information, did not use cross-series learning, and was not given a probabilistic loss function. This design was chosen to provide a transparent feasibility baseline — establishing whether raw sales history alone contains sufficient signal for a simple neural architecture to match established benchmarks — rather than to test modern neural forecasting at its current capability frontier. The result is therefore informative about the value of sales-history-only sequential modelling relative to feature-rich tree methods, not about the broader potential of neural forecasting for retail demand.

2.10 Research Gap

The reviewed literature provides strong foundations for large-scale retail forecasting, demand classification, gradient-boosted models, forecast evaluation and model interpretation. However, these areas are usually examined separately. Existing research does not provide a clear account of how forecasting performance and feature usefulness change across different forms of retail demand within one controlled experimental design.

The M5 competition demonstrated that machine-learning methods can outperform established benchmarks, but its aggregate results do not directly explain whether performance differences are associated with smooth, erratic, intermittent or lumpy demand (Makridakis, Spiliotis and Assimakopoulos, 2022). The ADI/CV² framework provides an established method for separating demand (Syntetos, Boylan and Croston, 2005), but the original study used short monthly automotive series and did not examine daily grocery demand or modern machine-learning models.

Previous studies have not established whether the same categories of retail information are equally valuable across different demand patterns. The explainability literature presents a related gap: SHAP provides theoretically grounded attribution (Lundberg and Lee, 2017; Lundberg et al., 2020), but these studies do not examine daily retail forecasting, compare explanations across ADI/CV² demand classes, or determine whether features receiving high SHAP importance improve out-of-sample accuracy.

This dissertation addresses these gaps through a demand-type-specific analysis of the Walmart M5 dataset. Its contribution is not the introduction of a new forecasting algorithm. Instead, it combines established methods within one consistent framework: ADI/CV² classification, four-model comparison (seasonal naive, Croston, SBA and XGBoost), within-type feature ablation, exact TreeSHAP analysis, and a restricted LSTM feasibility study.

2.11 Chapter Summary

This chapter reviewed the literature relevant to retail demand forecasting, demand heterogeneity, machine learning, feature analysis, explainability, hierarchical evaluation and deep-learning forecasting. The review showed that retail forecasting is not one uniform prediction problem and that forecasting difficulty changes according to the level of aggregation, the frequency of sales, the variability of non-zero demand and the information available at the forecast origin.

The central gap identified in the literature is the limited connection between demand-pattern classification, machine-learning performance, feature ablation and model explanation within one controlled hierarchical retail experiment. This dissertation addresses that gap by evaluating four forecasting models separately across the four demand types, testing five feature groups through ablation, comparing SHAP reliance with forecast benefit and conducting a limited LSTM feasibility analysis. The next chapter explains the research approach, data preparation, experimental design, forecasting models and evaluation procedures used to carry out this analysis.

Chapter 3: Research Approach

3.1 Research Design

This dissertation used a quantitative comparative forecasting design. The aim was not to develop a new forecasting algorithm, but to examine whether the performance and behaviour of established forecasting approaches differed across structurally different forms of retail demand. The item-store daily sales series in the Walmart M5 dataset formed the main unit of analysis.

The research design was based on four demand types: smooth, erratic, intermittent and lumpy. Each of the 30,490 bottom-level item-store series was classified using the Average Demand Interval and squared Coefficient of Variation. The classification was completed before the forecasting experiments so that model performance could be examined separately within each demand type.

A balanced experimental subset was used for model development and comparison. I selected 200 item-store series from each demand type, producing a total of 800 series. Equal group sizes were used to prevent the much larger intermittent and lumpy groups from dominating the comparison. This design made the demand-type results easier to compare, but it also means that the experimental sample does not represent the natural distribution of the complete M5 dataset.

The primary forecasting comparison involved four models. The seasonal naive model was used as the main benchmark. Croston's method and the Syntetos–Boylan Approximation (SBA) were included as demand-specific estimators from the intermittent-demand literature. XGBoost was used as the main machine-learning model. All models were evaluated using five expanding walk-forward folds with a fixed 28-day test window and the same series and forecast origins.

The feature-ablation analysis and exact TreeSHAP calculations were applied to the XGBoost models. A restricted sequence-only LSTM experiment was also conducted for smooth and erratic demand as a feasibility study. All main numerical comparisons were descriptive. Formal statistical significance was not claimed because the study contained only five independent forecast origins for each demand type.

3.2 Research Methodology Framework

This study follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which structures data science projects into six iterative phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment (Chapman et al., 2000). CRISP-DM was selected over alternative frameworks such as KDD (Knowledge Discovery in Databases) and SEMMA (Sample, Explore, Modify, Model, Assess) for three reasons. First, CRISP-DM explicitly separates business understanding from technical modelling, which maps directly to the distinction between demand-type classification and forecast evaluation in this study. Second, its iterative structure accommodates the feedback loops in this project — the classification stability audit in Section 3.4.1 and the denominator audit in Section 3.10 both arose from evaluation-phase findings that required revisiting the data preparation phase. Third, CRISP-DM is the most widely adopted framework in retail analytics and forecasting literature, making the methodological positioning transparent to practitioners.

Table 3.1 maps each CRISP-DM phase to the corresponding sections of this dissertation.

Table 3.1: CRISP-DM phase mapping

CRISP-DM Phase	Dissertation Section	Key Activities
Business Understanding	§1.1–§1.4	Define forecasting problem, research questions, scope
Data Understanding	§3.4–§3.6, §4.1–§4.3	M5 dataset exploration, demand classification, EDA
Data Preparation	§3.5–§3.8	Feature engineering, balanced sampling, fold construction
Modelling	§3.9–§3.14	XGBoost, Croston, SBA, LSTM implementation and tuning
Evaluation	§3.10, §4.4–§4.12	WRMSSE scoring, ablation, SHAP, stability audits
Deployment	§5.9, §6.3	Demand-type conditional framework, practical implications

3.3 Dataset and Study Population

The study used the Walmart M5 forecasting dataset. The dataset contains daily unit sales for products sold in ten Walmart stores across three US states: California, Texas and Wisconsin. There are 3,049 products and 30,490 item-store series in total. The available sales history covers 1,941 consecutive days, from 29 January 2011 to 22 May 2016, producing 59,181,090 item-store-day sales observations. Of these, 40,241,819 recorded zero sales, representing approximately 68.0% of all observations.

Three original M5 data sources were used: the sales file (daily unit sales), the calendar file (dates, events and SNAP schedules) and the selling-price file (weekly prices). These were combined to create one daily modelling table where each row represented one item-store series on one date.

3.3.1 Balanced Experimental Sample

A balanced subset was created containing 200 item-store series from each demand type (800 total). Each selected series retained its complete 1,941-day history, producing 1,552,800 item-store-day records. A fixed random seed of 42 was used for reproducible selection. The balanced subset was not intended to reproduce the natural composition of Walmart demand; results must be interpreted as demand-type comparisons rather than estimates of overall performance across the natural M5 distribution.

3.4 Demand Classification Procedure

The 30,490 item-store series were classified before forecasting models were developed, following the framework of Syntetos, Boylan and Croston (2005). ADI was calculated as the ratio of total daily observations to the number of positive-sales observations. CV^2 was calculated using only the positive-sales observations. The ADI threshold was 1.32 and the CV^2 threshold was 0.49.

Table 3.2: Demand-type classification results

Demand type	ADI rule	CV^2 rule	Series	Percentage
Smooth	$ADI \leq 1.32$	$CV^2 \leq 0.49$	983	3.224%
Erratic	$ADI \leq 1.32$	$CV^2 > 0.49$	494	1.620%
Intermittent	$ADI > 1.32$	$CV^2 \leq 0.49$	23,075	75.681%

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Demand type	ADI rule	CV ² rule	Series	Percentage
Lumpy	ADI > 1.32	CV ² > 0.49	5,938	19.475%
Total			30,490	100.000%

Intermittent and lumpy series together represented 95.156% of the complete item-store population. The classification was calculated once using the complete 1,941-day history and was not supplied to the forecasting models as a daily predictor.

Table 3.3: Mean demand characteristics in the balanced sample

Demand type	Mean zero-sales rate	Mean ADI	Mean CV ²
Smooth	14.382%	1.176	0.376
Erratic	16.827%	1.207	0.728
Intermittent	72.436%	7.234	0.286
Lumpy	59.889%	4.165	0.721

This sample composition is reproduced in Chapter 4 as Table 4.5 for reference when interpreting the demand-type-specific results.

3.4.1 Classification Stability Audit

The ADI/CV² classification used the complete 1,941-day sales history for each series. In a deployed forecasting system, demand type would need to be determined using only information available at the forecast origin. To assess whether the retrospective classification materially affected the study conclusions, a stability audit was conducted on all 800 selected series.

For each series, ADI and CV² were recalculated using only the training observations available before the earliest forecast origin — that is, sales from 2011-01-29 to 2016-01-03 (d_1 to $d_{1,801}$), covering 1,801 days. The resulting training-only classification was compared against the full-history classification used throughout the study.

Overall agreement was 93.8% (750 of 800 series). Agreement by demand type was: intermittent 98.5%, smooth 94.5%, lumpy 92.5%, and erratic 89.5%. Fifty series changed class when the final 140 days of history were removed.

Table 3.9: Classification stability confusion matrix (full-history versus training-only label)

Full-history label	Smooth (training)	Erratic (training)	Intermittent (training)	Lumpy (training)	Total
Smooth	189	2	9	0	200
Erratic	8	179	0	13	200
Intermittent	0	0	197	3	200
Lumpy	0	1	14	185	200
Total	197	182	220	201	800

Diagonal entries show series whose label was unchanged; off-diagonal entries show series whose training-only classification differed from the full-history label.

The dominant transition patterns were intermittent-to-lumpy or lumpy-to-intermittent (17 series crossing the CV² = 0.49 boundary) and erratic-to-lumpy or lumpy-to-erratic

(14 series crossing the $ADI = 1.32$ boundary). Intermittent demand was the most stable class; erratic demand was the least stable.

Of the 800 series, 415 fell within 0.10 of at least one classification threshold (199 near $ADI = 1.32$, 309 near $CV^2 = 0.49$). These borderline series are the primary source of instability and represent a genuine limitation of applying fixed cut-offs to daily retail data with gradual demand evolution.

The stability audit indicates that the retrospective classification was broadly stable. The study's strongest finding — that Croston and SBA outperformed XGBoost for intermittent and lumpy demand — rests on the two most stable demand classes (98.5% and 92.5% agreement respectively). Results for erratic demand, where agreement was lowest at 89.5%, should be interpreted with greater caution, as a portion of erratic series may have been classified differently had training-only history been used from the outset. All results throughout this dissertation reflect the original full-history classification; the training-only classification is reported here as a sensitivity check only.

3.5 Data Preparation and Feature Engineering

The original M5 files were transformed into a daily supervised-learning dataset. The sales data were converted from wide to long format, joined with calendar and selling-price information, and augmented with historical sales features. After applying the 56-day historical look-back requirement, the balanced sample contained 1,508,000 feature-complete rows.

Table 3.4 provides a complete dictionary of all 18 predictors used in the XGBoost models, including exact feature names, transformation method, information availability at forecast time, and missing-value treatment.

Table 3.4: XGBoost feature dictionary

Feature name	Group	Transformation	Lag / window	Available at forecast time	Missing-value treatment
lag_7	Lag	Raw sales, shifted 7 days	7 days	Yes (historical)	Native XGBoost handling
lag_14	Lag	Raw sales, shifted 14 days	14 days	Yes (historical)	Native XGBoost handling
lag_28	Lag	Raw sales, shifted 28 days	28 days	Yes (historical)	Native XGBoost handling
lag_56	Lag	Raw sales, shifted 56 days	56 days	Yes (historical)	Native XGBoost handling
rolling_mean_7	Rolling	Mean of prior 7 days sales	7 days	Yes (historical)	Native XGBoost handling
rolling_std_7	Rolling	Std dev of prior 7 days sales	7 days	Yes (historical)	Native XGBoost handling
rolling_mean_28	Rolling	Mean of prior 28 days sales	28 days	Yes (historical)	Native XGBoost handling
rolling_std_28	Rolling	Std dev of prior 28 days sales	28 days	Yes (historical)	Native XGBoost handling
is_weekend	Calendar	Binary: 1 if Saturday or Sunday	—	Yes (known future)	None (complete)
weekday	Calendar	Integer 0–6 (Monday=0)	—	Yes (known future)	None (complete)

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Feature name	Group	Transformation	Lag / window	Available at forecast time	Missing-value treatment
month	Calendar	Integer 1–12	—	Yes (known future)	None (complete)
year	Calendar	Integer (e.g. 2015, 2016)	—	Yes (known future)	None (complete)
event_flag	Event/SNAP	Binary: 1 if named M5 event	—	Yes (known future)	None (complete)
event_type_encoded	Event/SNAP	Integer encoding of event type	—	Yes (known future)	None (complete)
snap_flag	Event/SNAP	Binary: 1 if SNAP active in state	—	Yes (known future)	None (complete)
sell_price	Price	Raw weekly selling price	—	Yes (known future)	Retained as missing; native XGBoost handling
price_change	Price	sell_price minus previous week price	—	Yes (known future)	Retained as missing; native XGBoost handling
price_vs_28d_avg	Price	sell_price minus 28-day rolling average price	—	Yes (known future)	Retained as missing; native XGBoost handling

The final XGBoost dataset contained 18 predictors organised into five groups:

Table 3.5: XGBoost feature structure

Feature group	Number of features	Information represented
Lag	4	Sales at selected earlier dates
Rolling	4	Recent demand level and variation
Calendar	4	Recurring date and time structure
Event/SNAP	3	Named events and SNAP schedules
Price	3	Current and relative selling price
Total	18	—

Missing selling-price rates were 0.642% for smooth, 0.399% for erratic, 20.742% for intermittent and 18.890% for lumpy demand. Missing prices were retained rather than imputed; XGBoost handled them natively within its tree-building process. All historical features were shifted before calculation to prevent target leakage, and known future information (calendar, events, SNAP, prices) could be supplied for the forecast horizon because these values were not derived from future sales.

3.6 Walk-Forward Validation and Forecasting Design

The forecasting experiment used an expanding walk-forward validation design with five outer forecast origins. At each origin, the model was trained using only observations occurring before the start of the test period; the next 28 consecutive days formed the

outer test window. The same five forecast origins and 28-day test windows were used for all demand types and forecasting models.

For the balanced sample, each fold contained 22,400 bottom-level forecast observations (800 series $\times$ 28 days). Across all five folds, each complete forecasting method generated 112,000 bottom-level forecasts. Within each demand type, one fold contained 5,600 forecast observations (200 series $\times$ 28 days), yielding 28,000 out-of-sample forecasts per demand type per model.

Bergmeir, Hyndman and Koo (2018) show that standard random k-fold cross-validation can produce valid error estimates for autoregressive forecasting models when the model's residuals are uncorrelated, which is more likely when a flexible model, such as a machine learning method, is fit against a simpler underlying process. This complicates the common assumption that time-ordered validation is always necessary for time series problems. However, this project used walk-forward validation with five expanding folds rather than random k-fold splitting. This was a deliberate, conservative choice: verifying that XGBoost, Croston and SBA residuals meet the uncorrelated-error condition specified by Bergmeir et al. across four demand types would require additional diagnostic testing outside the scope of this study, and walk-forward validation avoids relying on that assumption altogether by preserving temporal order regardless of whether it holds.

3.7 Seasonal Naive Benchmark

The seasonal naive model repeated the final seven observed training days across each 28-day test horizon. This provided a simple, transparent benchmark based on weekly retail seasonality that required no parameter estimation. The complete 28-day forecast was generated from one fixed origin; actual sales observed during the test period were not used to update later forecasts.

3.8 Croston's Method and SBA

Croston's method (Croston, 1972) was applied to all four demand types using the same five expanding walk-forward folds and 28-day test windows. For each item-store series and fold, the method decomposed the training history into two components: the non-zero demand sizes and the inter-demand intervals. Each component was updated using simple exponential smoothing with a smoothing parameter of 0.1. The flat-rate forecast for each test day was obtained by dividing the smoothed demand-size estimate by the smoothed inter-demand interval estimate.

The Syntetos–Boylan Approximation (Syntetos and Boylan, 2001; 2005) applied a bias-correction factor to the Croston forecast. Both methods produced a constant (flat) forecast across the 28-day horizon because they do not incorporate calendar, event or price information. They were evaluated using the same WRMSSE, MAE and RMSE calculations as the seasonal naive and XGBoost models, with forecasts aggregated across the same reconstructed hierarchy.

Including Croston and SBA alongside the seasonal naive benchmark and XGBoost created a four-model comparison. This allowed the study to test whether demand-specific estimators designed for irregular demand provided forecasting value in a daily retail context and whether their performance pattern differed from the weekly-repetition benchmark and the feature-based machine-learning model.

Teunter, Syntetos and Babai (2011) propose an alternative intermittent-demand method, TSB, which updates the probability of demand occurring in every period rather than only after a positive demand event, as Croston and SBA do. This allows TSB to detect declining or obsolescent demand more quickly: a run of zero-demand periods

steadily lowers the estimated occurrence probability, whereas Croston and SBA remain anchored to the last observed positive demand until another sale occurs. TSB was not implemented in this study. Its inclusion would be particularly relevant for the lumpy demand group, where XGBoost's fold-to-fold WRMSSE variability was highest of any model-demand combination tested (standard deviation 0.2721), which may partly reflect underlying demand-regime changes that Croston and SBA, by design, are slower to detect. This is noted as a specific direction for future work.

3.8.1 Smoothing-Parameter Sensitivity

I fixed the smoothing parameter at $\alpha = 0.10$ for both Croston and SBA, following the convention used in the intermittent-demand literature rather than optimising it through cross-validation. To check whether this choice affected the main conclusions, I tested α values across 0.05 to 0.20 for both methods.

For Croston on lumpy demand, $\alpha = 0.10$ was optimal, with mean WRMSSE ranging from 1.0674 (at $\alpha = 0.10$) to 1.0858 across the tested range. For SBA on intermittent demand, $\alpha = 0.10$ was not optimal: both $\alpha = 0.15$ and $\alpha = 0.20$ produced lower mean WRMSSE than the selected value. This means the reported SBA intermittent result is conservative—a tuned smoothing parameter would have produced a stronger result for SBA.

The headline finding that Croston and SBA outperformed XGBoost for intermittent and lumpy demand held across the entire tested α range. The sensitivity to α is nonetheless a genuine limitation: cross-validated smoothing-parameter selection could have improved the Croston and SBA results, particularly for SBA on intermittent demand.

3.9 XGBoost Model Development

XGBoost was selected as the primary machine-learning model rather than LightGBM or CatBoost for three reasons. First, the M5 competition evidence showed that gradient-boosted tree methods — including both XGBoost and LightGBM — dominated the top submissions, with no decisive accuracy advantage established between the two frameworks at the item-store level (Makridakis, Spiliotis and Assimakopoulos, 2022). Second, XGBoost's native TreeSHAP integration allowed exact SHAP values to be calculated using the model's internal tree structure, which was essential for the SHAP-versus-ablation comparison that forms a core contribution of this study. Third, XGBoost's reg:tweedie objective with a configurable variance power parameter was a natural fit for the non-negative, zero-heavy, right-skewed daily retail sales distribution. LightGBM offers faster training on large datasets, but the computational difference was not a limiting factor for the 800-series experimental sample used here.

The Tweedie objective was selected because daily retail sales at the item-store level are non-negative, contain many exact zeros, and exhibit right-skewed positive-demand distributions — properties consistent with a compound Poisson–gamma generative process. A Tweedie variance power of 1.5 was used, placing the model between a pure Poisson distribution (power = 1) and a pure gamma distribution (power = 2). This choice follows the distributional reasoning in Zhou, Qian and Yang (2022) and was not formally optimised through a distributional comparison; alternative variance power values are noted as a direction for future work.

Recursive multi-step forecasting was used rather than a direct multi-horizon strategy. Under the recursive approach, the model first forecast day one of the 28-day horizon; that prediction was inserted into the evolving feature history; and lag and rolling features were recalculated for subsequent days. Known future information — calendar

dates, events, SNAP schedules and selling prices — was supplied directly for the full horizon. The recursive design allowed the complete feature set to remain available throughout the forecast horizon, but it also introduced error propagation: forecast errors on early days feed into the lag and rolling features used for later days. Recursive forecasting was considered a plausible contributor to weaker XGBoost performance on sparse demand; however, the horizon-day error analysis in Section 4.13 found that error follows a weekly cyclical pattern rather than a monotonic increase across the horizon, suggesting that day-of-week demand structure is a stronger driver than recursive propagation.

The balanced experimental sample of 200 series per demand type was a deliberate analytical design choice rather than a representative sampling strategy. Equal group sizes ensured that each demand type contributed equally to every model comparison, preventing the much larger intermittent group (23,075 series, 75.7% of the population) from dominating the results. This design made within-type and cross-type comparisons directly interpretable, but it means the experimental results do not represent the natural distribution of Walmart demand. Sampling fractions ranged from 0.87% for intermittent to 40.49% for erratic demand (Table C.1). All results should be interpreted as demand-type-specific evidence from the selected sample, not as estimates of overall M5 population performance.

A separate global XGBoost model was trained for each demand type and outer validation fold (4 demand types $\times$ 5 folds = 20 final models). The models used the reg:tweedie objective with a Tweedie variance power of 1.5. The number of boosting rounds was selected separately for every demand type and fold using an internal 28-day validation period and recursive WRMSSE. After selection, the final model was refitted using the complete outer training period.

Table 3.6 reports the full XGBoost hyperparameter specification. All parameters not listed were retained at their XGBoost 1.7 software defaults.

Table 3.6: XGBoost hyperparameter specification

Parameter	Value	Notes
objective	reg:tweedie	Non-negative zero-heavy target
tweedie_variance_power	1.5	Mid-range compound Poisson–gamma
eta (learning rate)	0.05	Fixed across all models
max_depth	6	Fixed across all models
min_child_weight	1.0	Software default
subsample	0.8	Row subsampling per tree
colsample_bytree	0.8	Feature subsampling per tree
gamma	0.0	Software default
reg_alpha	0.0	Software default
reg_lambda	1.0	Software default
max_bin	256	Histogram bins
tree_method	hist	Histogram-based construction
seed	42	Fixed for reproducibility
Boosting rounds	Selected per fold and demand type	Grid search over recursive WRMSSE; candidate rounds evaluated at checkpoints (25–1,500); best checkpoint refined ± 50 in steps of 10; final model

Parameter	Value	Notes
		retrained from scratch on full outer-training period

XGBoost generated the 28-day horizon recursively: the model first forecast day one, that prediction was inserted into the evolving sales history, and lag and rolling features were recalculated for subsequent days. Known future information (calendar, events, SNAP, prices) was supplied directly. Actual sales from the outer test period were never inserted into the feature history during this process.

3.10 Forecast Evaluation Metrics and Hierarchy Reconstruction

Forecasts were generated at the item-store level only. The same bottom-level predictions were summed upward to construct all 12 hierarchy levels, so forecasts were automatically coherent across every level and no reconciliation step was required. For each hierarchy series, RMSSE scales the model's squared forecast error by the mean squared one-step naive error calculated from the in-sample training period. Revenue weights were calculated from the dollar sales recorded in the final 28 training days of each fold, using only observations inside the outer training window. These weights and scaling denominators were held constant across all four models within each fold, so every model comparison used identical scaling. The final WRMSSE score averaged the scaled errors across all 12 hierarchy levels with equal weight given to each level — meaning each of the 12 aggregation levels contributed equally to the final score, not each individual series. The WRMSSE denominator audit confirmed that the maximum absolute difference between any pair of scaling denominators for the same series and fold was 0.0 across all 90 model-demand-type-fold combinations, confirming identical scaling throughout. This implementation differs from the official M5 competition evaluation in two respects: the hierarchy was reconstructed within the balanced subset of 800 series rather than the full 30,490-series population, and revenue weights were recalculated within each demand-type subset rather than using the official competition weights. Results therefore reflect within-group forecasting performance for the selected series and should not be compared directly with official M5 leaderboard scores.

MAE and RMSE were retained as secondary bottom-level measures. Model improvement was calculated as the percentage reduction in mean WRMSSE relative to the seasonal naive benchmark, and fold-level wins were counted. No formal significance test was applied because the five outer forecast origins were insufficient for reliable statistical inference.

Hierarchical forecasting research addresses the problem of forecast incoherence: when different levels of an aggregation hierarchy are forecast independently, the resulting predictions typically do not sum correctly across levels, requiring reconciliation techniques to adjust base forecasts into a coherent structure (Petropoulos et al., 2022). This study avoids that problem by construction. All four forecasting methods generate predictions at the item-store level only; the twelve-level WRMSSE hierarchy used for evaluation is produced by summing these same bottom-level forecasts upward, rather than forecasting each level independently. The forecasts are therefore automatically coherent across all levels, and no reconciliation step was required.

3.11 Feature-Ablation Methodology

A leave-one-feature-group-out ablation study examined the conditional forecasting contribution of each feature group. One complete group was removed at a time, and the corresponding XGBoost model was retrained using the remaining predictors while retaining the boosting rounds selected for the full model. This produced 100 ablation models (5 groups $\times$ 4 demand types $\times$ 5 folds) and 560,000 bottom-level forecasts. The ablation measure was $\Delta\text{WRMSSE} = \text{WRMSSE}_{\text{ablated}} - \text{WRMSSE}_{\text{full}}$.

3.12 SHAP Implementation

Exact native TreeSHAP values were calculated for all 112,000 final XGBoost forecasts. The SHAP audit confirmed that the maximum difference between the raw XGBoost margin and the value reconstructed from SHAP contributions was approximately 2.19×10^{-5} , within normal floating-point precision. This figure represents the maximum absolute residual across all predictions; the mean residual was effectively zero, consistent with the exact additivity property of TreeSHAP reported in Section 4.9. Absolute SHAP values were aggregated by demand type and feature group to produce normalised reliance shares.

3.13 LSTM Feasibility Design

The LSTM experiment was restricted to smooth and erratic demand. The sequence-only network used a 56-day input sequence, 32 LSTM units, a 16-unit dense layer and a direct 28-day output (5,356 trainable parameters). It did not receive calendar, event, SNAP, price or engineered lag/rolling features. Ten final LSTM models (2 demand types $\times$ 5 folds) produced 56,000 forecasts in approximately 530.58 minutes in the CPU-based environment.

Table 3.7 provides the complete LSTM architecture and training specification.

Table 3.7: LSTM feasibility experiment specification

Parameter	Value
Input length	56 days (univariate sales history only)
Output horizon	28 days (direct multi-output)
Architecture	Input(56,1) $\rightarrow$ LSTM(32 units, tanh activation) $\rightarrow$ Dense(16 units, ReLU) $\rightarrow$ Dense(28 units, linear)
Total trainable parameters	5,356
Optimiser	Adam
Learning rate	0.001
Gradient clipping	clipnorm = 1.0
Loss function	Mean squared error (MSE)
Batch size	512
Maximum epochs	50
Early stopping	Monitored val_loss; patience = 5; min_delta = 1×10^{-4} ; restore best weights
Input scaling	log1p transform followed by per-series z-score normalisation (mean and std from training window only; ddof = 0)
Covariates	None — sales history only; no calendar, event, SNAP or price features
Refit procedure	Epoch count selected on inner validation; model rebuilt from scratch and retrained for exactly that many epochs on full outer-training period without early stopping

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Parameter	Value
Post-processing	Negative forecasts clipped to zero after inverse transform
Random seed	42 (with TF_DETERMINISTIC_OPS=1 and enable_op_determinism())
Models fitted	10 (2 demand types × 5 folds)
Runtime environment	CPU only; total runtime approximately 530.58 minutes

3.14 WRMSSE Denominator Audit

A separate denominator audit checked all WRMSSE scaling denominators across the 90 model–demand-type–fold combinations (seasonal naive, Croston, SBA and XGBoost × 4 demand types × 5 folds, plus the LSTM combinations). The maximum absolute difference between any pair of denominators for the same series and fold was 0.0, confirming that all models were evaluated using identical scaling.

3.15 Ethical Considerations and Reproducibility

Table 3.8 provides a compact experiment specification covering the key design parameters required to reproduce the primary analysis.

Table 3.8: Experiment specification summary

Parameter	Value
Data source	Walmart M5 (sales, calendar, sell_prices files)
Data period	2011-01-29 to 2016-05-22 (d_1 to d_1,941)
Series used	800 (200 per demand type)
Random seed	42
Fold 1 test window	2016-01-04 to 2016-01-31
Fold 2 test window	2016-02-01 to 2016-02-28
Fold 3 test window	2016-02-29 to 2016-03-27
Fold 4 test window	2016-03-28 to 2016-04-24
Fold 5 test window	2016-04-25 to 2016-05-22
Models compared	Seasonal naive, Croston, SBA, XGBoost
Total XGBoost models fitted	20 (4 demand types × 5 folds)
Total final XGBoost forecasts	112,000 (800 series × 28 days × 5 folds)
Forecasts per demand-type/fold model	5,600 (200 series × 28 days)
Ablation models fitted	100 (5 feature groups × 4 demand types × 5 folds)
LSTM models fitted	10 (2 demand types × 5 folds)
Primary metric	WRMSSE
Software	Python 3.x; XGBoost 1.7; TensorFlow/Keras

The M5 data are publicly available and contain no personally identifiable information. A fixed random seed of 42 was used throughout. Intermediate datasets, forecasts, metrics and audit outputs were saved at each stage. The same locked evaluation implementation was reused for all models. The study did not involve human participants, personal data or commercial decisions. The reproducibility claim in this study is computational reproducibility from saved forecast outputs and model

parameters, not full end-to-end retraining reproducibility from raw M5 files. The complete codebase, including all notebooks, fold definitions, feature engineering scripts, model configurations and saved outputs, is publicly available at https://github.com/rishabrijk/MSc_Dissertation_M5 (Kothari, 2026).

3.16 Chapter Summary

This chapter described the full methodological pipeline used in the study, structured around the CRISP-DM framework introduced in Section 3.2. The Walmart M5 dataset was classified into four demand types using the ADI/CV² framework, and a balanced experimental sample of 200 series per demand type was constructed for controlled comparison. Feature engineering produced 18 predictors across five groups, all verified as available at forecast time. Four models were implemented — seasonal naive, Croston, SBA and XGBoost — and evaluated using five expanding walk-forward folds with a fixed 28-day test horizon. WRMSSE was adopted as the primary metric, with a full denominator audit confirming identical scaling across all model comparisons. A restricted LSTM feasibility experiment was added for smooth and erratic demand. Reproducibility was addressed through saved forecast outputs, fixed random seeds, documented fold dates and the experiment specification in Table 3.7. The classification stability audit in Section 3.4.1 confirmed that 93.8% of series retained their demand-type label under training-only classification, supporting the validity of the demand-type-specific evaluation design.

Chapter 4: Data Analysis and Results

4.1 Chapter Overview

This chapter presents the empirical results from the demand classification, exploratory analysis, seasonal naive benchmark, Croston and SBA evaluation, XGBoost forecasting experiment, four-model comparison, hierarchy-level evaluation, feature-ablation study, SHAP analysis and LSTM feasibility experiment. WRMSSE is the primary metric; lower values indicate better forecasting performance.

4.2 Demand Classification and Exploratory Findings

4.2.1 Distribution of Demand Types

The full M5 item-store population contained 30,490 series. The ADI/CV² classification produced the distribution shown in Table 4.1.

Table 4.1: Demand-type distribution across the complete M5 item-store population

Demand type	Number of series	Percentage
Smooth	983	3.224%
Erratic	494	1.620%
Intermittent	23,075	75.681%
Lumpy	5,938	19.475%
Total	30,490	100.000%

Intermittent and lumpy series together represented 95.156% of the complete population. The complete dataset contained 59,181,090 item-store-day observations, of which 40,241,819 (approximately 68.0%) recorded zero sales.

4.2.2 Exploratory Data Analysis

Figure 4.1 shows the 28-day rolling average of total daily sales across the full M5 dataset. The series displays clear annual seasonality, a long-term upward trend and several sharp spikes associated with major retail events.

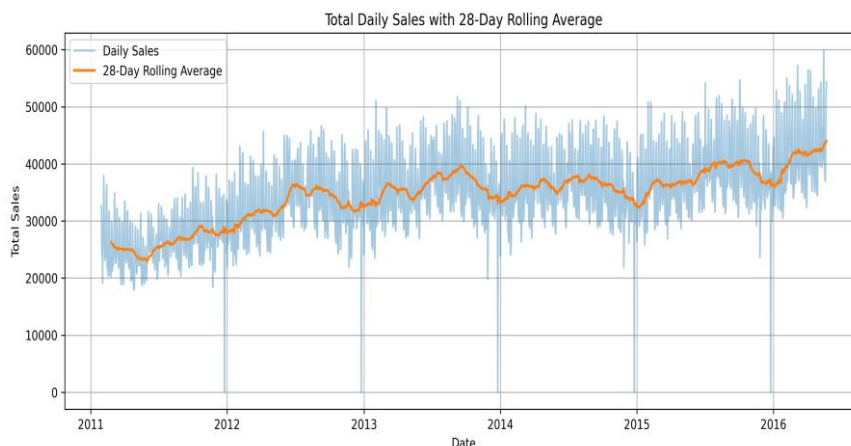


Figure 4.1: Total daily sales with 28-day rolling average

RETAIL DEMAND FORECASTING UNDER DEMAND HETEROGENEITY

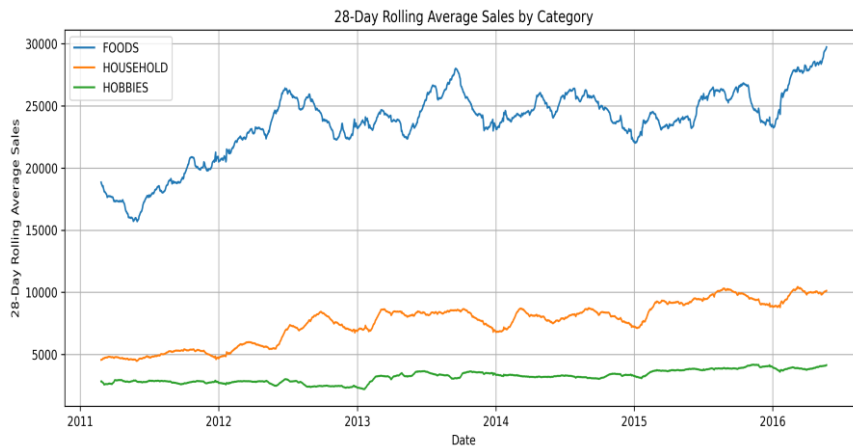


Figure 4.2: Category-level sales with 28-day rolling average

Table 4.2 summarises total sales by product category. The FOODS category accounted for the largest share of total unit sales.

Table 4.2: Total sales by product category

Category	Total sales (units)
FOODS	45,922,427
HOUSEHOLD	14,764,090
HOBBIES	6,240,656

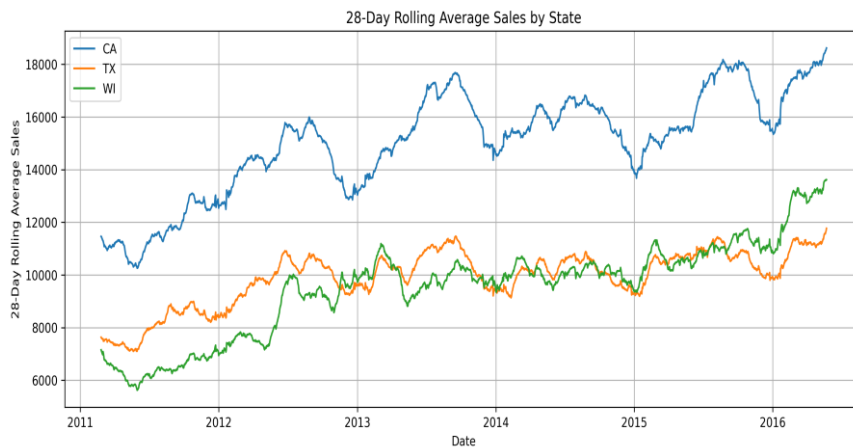


Figure 4.3: State-level sales with 28-day rolling average

Table 4.3: Total sales by state

State	Total sales (units)
California	29,196,717
Texas	19,228,405
Wisconsin	18,502,051

RETAIL DEMAND FORECASTING UNDER DEMAND HETEROGENEITY

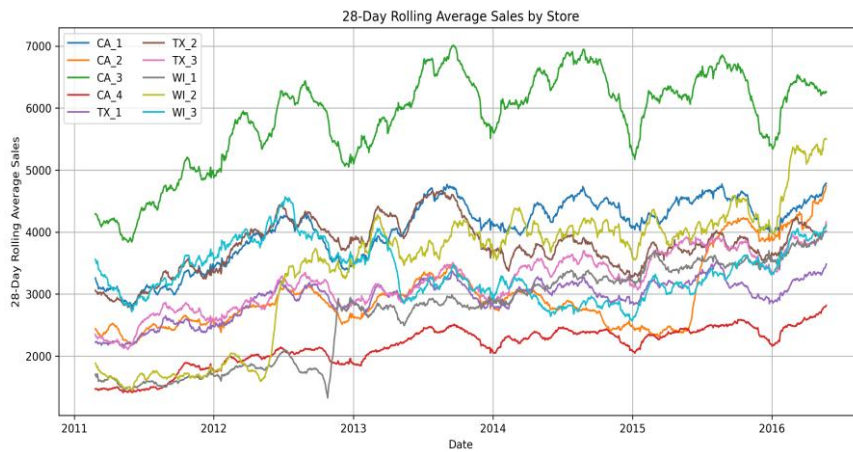


Figure 4.4: Store-level sales with 28-day rolling average

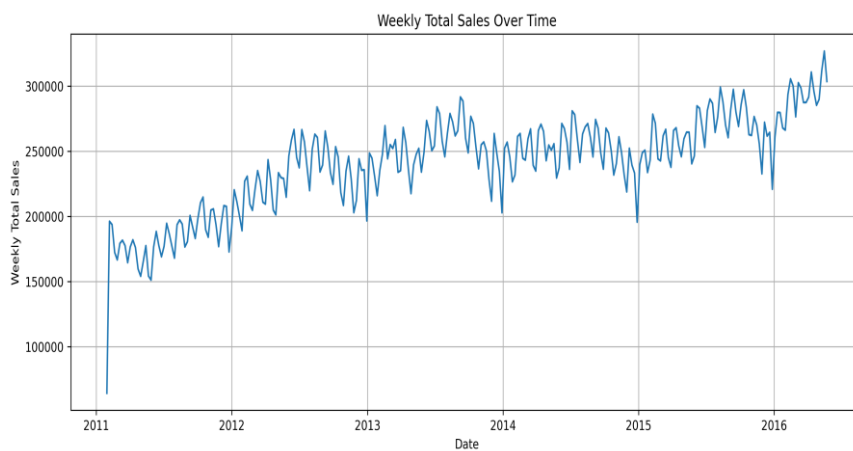


Figure 4.5: Weekly total sales pattern

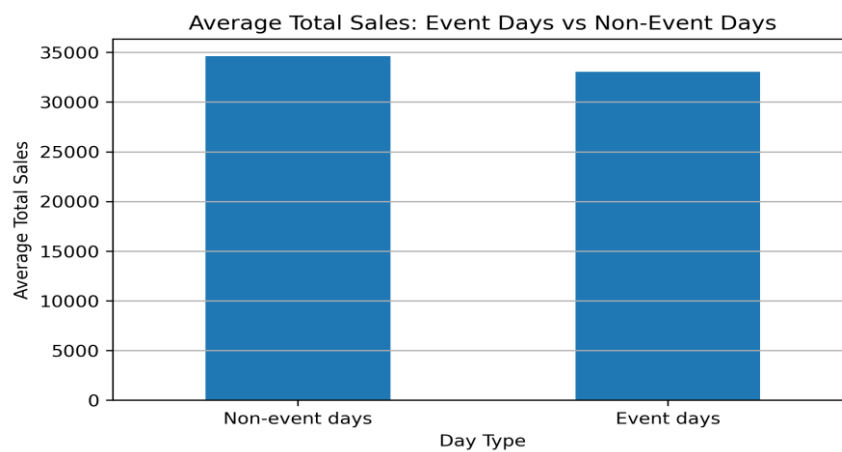


Figure 4.6: Event versus non-event mean daily sales

RETAIL DEMAND FORECASTING UNDER DEMAND HETEROGENEITY

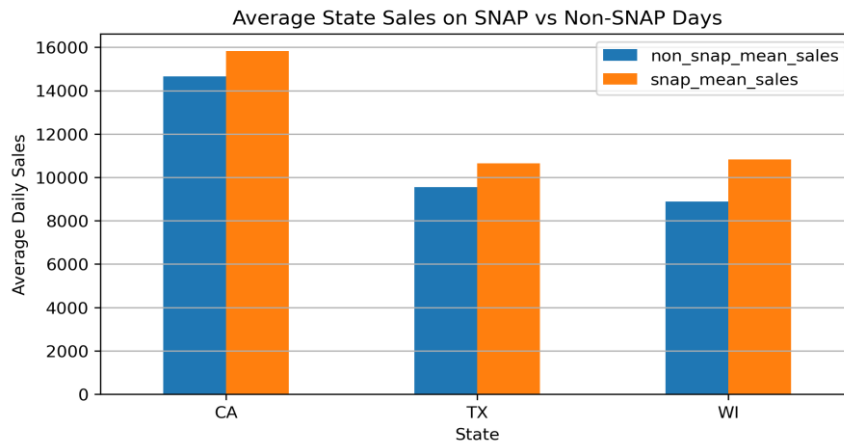


Figure 4.7: SNAP versus non-SNAP mean daily sales by state

Table 4.4: SNAP sales effect by state

State	Non-SNAP mean sales	SNAP mean sales	Percentage difference
California	14,657.74	15,823.42	+7.95%
Texas	9,543.86	10,643.51	+11.52%
Wisconsin	8,892.50	10,832.66	+21.82%

4.2.3 Demand-Type Characteristics

Table 4.5: Mean demand characteristics in the balanced experimental sample

Demand type	Mean zero-sales rate	Mean ADI	Mean CV ²
Smooth	14.382%	1.176	0.376
Erratic	16.827%	1.207	0.728
Intermittent	72.436%	7.234	0.286
Lumpy	59.889%	4.165	0.721

This table reproduces Table 3.3 from Chapter 3 for convenience.

4.3 Seasonal Naive Benchmark Results

Table 4.6: Seasonal naive performance by demand type

Demand type	Mean WRMSSE	SD	Mean MAE
Smooth	0.8064	0.0924	3.3350
Erratic	0.9500	0.1603	4.4505
Intermittent	1.2156	0.2362	0.8940
Lumpy	1.2589	0.0665	1.7669

The seasonal naive model produced its lowest WRMSSE for smooth demand (0.8064) and its highest for lumpy demand (1.2589). Intermittent demand recorded a low MAE of 0.8940 despite a high WRMSSE of 1.2156, illustrating that a low average unit error can be misleading when most observations are zero.

4.4 Croston and SBA Results

Croston's method and SBA were evaluated using the same 800 series, five forecast origins and 28-day test periods. Table 4.7 presents the mean WRMSSE across the five outer folds for all four models.

Table 4.7: Mean WRMSSE by model and demand type (five-fold average)

Demand type	Seasonal naive	Croston	SBA	XGBoost
Smooth	0.8064	0.8556	0.8536	0.6360
Erratic	0.9500	0.8819	0.8853	0.7938
Intermittent	1.2156	1.1156	1.0978	1.2095
Lumpy	1.2589	1.0674	1.0756	1.1275

The four-model comparison produced a different pattern for each demand type. For smooth demand, both Croston (0.8556) and SBA (0.8536) produced higher WRMSSE than the seasonal naive benchmark (0.8064). XGBoost (0.6360) achieved the lowest score by a substantial margin. The flat-rate Croston and SBA forecasts could not capture the weekly seasonality that the seasonal naive benchmark reproduced directly.

For erratic demand, the pattern reversed for Croston and SBA relative to the benchmark. Croston (0.8819) and SBA (0.8853) both improved on the seasonal naive score of 0.9500. This matters because erratic demand has an ADI below 1.32—these series sell frequently. The improvement suggests that Croston and SBA helped not because they handled long zero intervals, but because their smoothed demand-level estimate reduced the effect of high positive-demand variability ($CV^2 > 0.49$) that made the seasonal naive's weekly repetition unstable. XGBoost (0.7938) still achieved the lowest WRMSSE for erratic demand.

For intermittent demand, Croston (1.1156) and SBA (1.0978) both improved substantially on the seasonal naive score of 1.2156. SBA recorded the lowest WRMSSE among all four models for intermittent demand, outperforming XGBoost (1.2095). This result is consistent with the original design purpose of these methods: intermittent series have ADI above 1.32 and comparatively stable positive quantities, making the smoothed demand-level approach well suited to this demand structure.

For lumpy demand, Croston (1.0674) and SBA (1.0756) again improved on the seasonal naive score of 1.2589. Both also outperformed XGBoost (1.1275) for lumpy demand. Lumpy series combine infrequent demand with variable positive quantities; Croston and SBA do not use recursive lag updating and therefore avoid the weekly error cycling observed in XGBoost (Section 4.13), though the horizon-day analysis suggests this cycling reflects demand periodicity rather than pure propagation error.

The Croston and SBA results therefore show a consistent pattern: these methods helped wherever demand deviated from smooth regularity—via high demand-size variability (erratic), via sparsity (intermittent), or via both (lumpy). Their only weak point was smooth demand, where the seasonal naive's weekly repetition was already effective and the flat-rate forecast could not represent the regular weekly structure.

4.5 XGBoost Forecasting Results

4.5.1 Overall WRMSSE Performance

Table 4.8: Seasonal naive versus XGBoost WRMSSE by demand type

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Demand type	Seasonal naive	XGBoost	Improvement	Fold wins
Smooth	0.8064	0.6360	21.13%	5/5
Erratic	0.9500	0.7938	16.44%	5/5
Intermittent	1.2156	1.2095	0.50%	4/5
Lumpy	1.2589	1.1275	10.44%	4/5

XGBoost improved mean WRMSSE for all four demand types relative to the seasonal naive benchmark. The improvement ranged from 21.13% for smooth demand to only 0.50% for intermittent demand. XGBoost won all five folds for smooth and erratic demand, and four of five for intermittent and lumpy demand (both losing during Fold 1). Across all 20 demand-type-fold comparisons, XGBoost achieved 18 WRMSSE wins, 20 MAE wins and 20 RMSE wins. The difference between WRMSSE and the secondary metrics shows that bottom-level unit-error improvement did not always translate into improvement across the scaled and weighted hierarchy.

4.5.2 Fold-Level Variation

Fold 4 was XGBoost's strongest fold across all four demand types simultaneously: smooth 0.5389, erratic 0.6448, intermittent 0.9247, lumpy 0.8492. These were the lowest WRMSSE values XGBoost achieved in any fold, for every demand type. No other fold produced the best XGBoost result for more than one demand type.

A separate finding involves Fold 1. In the seasonal naive versus XGBoost comparison, both WRMSSE losses for XGBoost—intermittent and lumpy—occurred in Fold 1. XGBoost performed anomalously weak in that fold relative to its average across the other four folds. This is a different observation from the Fold 4 result: Fold 4 was an instance of XGBoost performing anomalously strong (and in a different fold, against different benchmarks in the four-model comparison with Croston and SBA), while Fold 1 was an instance of XGBoost performing anomalously weak in the two-model naive comparison. The two findings involve different folds, different comparisons and different directions of deviation.

4.6 Four-Model Comparison

Table 4.9 presents the complete four-model comparison, showing the best-performing model for each demand type according to mean WRMSSE.

Table 4.9: Best model by demand type (lowest mean WRMSSE)

Demand type	Best model	Best WRMSSE	Second best	Second WRMSSE
Smooth	XGBoost	0.6360	Seasonal naive	0.8064
Erratic	XGBoost	0.7938	Croston	0.8819
Intermittent	SBA	1.0978	Croston	1.1156
Lumpy	Croston	1.0674	SBA	1.0756

Table 4.10 reports fold-level variability of WRMSSE across the five cross-validation folds, by model and demand type.

Table 4.10: Fold-level variability of WRMSSE by model and demand type

RETAIL DEMAND FORECASTING UNDER DEMAND HETEROGENEITY

Model	Demand type	Mean WRMSSE	Std Dev	Min	Max	CV%
Naive	Smooth	0.8064	0.0924	0.7197	0.9394	11.5%
Naive	Erratic	0.9500	0.1603	0.8296	1.2000	16.9%
Naive	Intermittent	1.2156	0.2362	0.9947	1.5583	19.4%
Naive	Lumpy	1.2589	0.0665	1.1687	1.3338	5.3%
XGBoost	Smooth	0.6360	0.0756	0.5389	0.7501	11.9%
XGBoost	Erratic	0.7938	0.1595	0.6448	1.0537	20.1%
XGBoost	Intermittent	1.2095	0.2291	0.9247	1.4720	18.9%
XGBoost	Lumpy	1.1275	0.2721	0.8492	1.5329	24.1%
Croston	Smooth	0.8556	0.0424	0.8106	0.8977	5.0%
Croston	Erratic	0.8819	0.0891	0.7907	0.9781	10.1%
Croston	Intermittent	1.1156	0.1346	0.9628	1.2580	12.1%
Croston	Lumpy	1.0674	0.0651	1.0210	1.1815	6.1%
SBA	Smooth	0.8536	0.0694	0.7616	0.9380	8.1%
SBA	Erratic	0.8853	0.1164	0.7900	1.0159	13.2%
SBA	Intermittent	1.0978	0.1477	0.9382	1.2716	13.5%
SBA	Lumpy	1.0756	0.0790	1.0036	1.2079	7.3%

XGBoost's WRMSSE for lumpy demand shows the highest variability of any model-demand combination ($CV\% = 24.1\%$), swinging from 0.8492 to 1.5329 across folds. This indicates that while Croston achieves the lowest mean WRMSSE for lumpy demand, XGBoost occasionally outperforms it in a given fold; the mean advantage for Croston is not uniform across all five test periods. By contrast, Croston and SBA show consistently lower variability than XGBoost and the naive baseline across nearly every demand type, suggesting these methods are not only more accurate on average for intermittent and lumpy demand but also more reliably so.

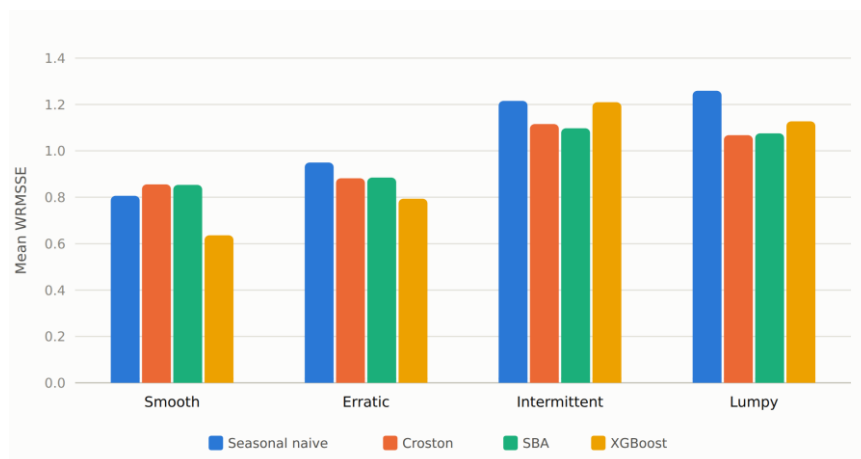


Figure 4.8: Mean WRMSSE by model and demand type (five-fold average). Lower values indicate better forecasting performance.

The four-model comparison yields a clear demand-type-specific pattern. XGBoost was the strongest model for smooth and erratic demand, where frequent observations provided a continuous signal for the 18-feature recursive design. For intermittent and lumpy demand, the simpler Croston and SBA methods outperformed both XGBoost

and the seasonal naive benchmark. This pattern supports the central thesis: model suitability depends on demand structure, and no single model dominated across all four types.

4.7 Hierarchy-Level Forecasting Results

The hierarchy breakdown outputs are available for Croston and SBA across intermittent and lumpy demand only. For these two models and demand types, WRMSSE increased consistently from aggregated levels to item-store level — a pattern expected under bottom-up aggregation, where aggregation smooths idiosyncratic series-level errors. Separate hierarchy breakdown files were not saved for XGBoost and the seasonal naive baseline during the original pipeline run, so a complete four-model hierarchy comparison table cannot be constructed without rerunning those models. The qualitative pattern — that all models perform better at aggregated levels than at item-store level — is consistent with the M5 competition findings reported in Makridakis, Spiliotis and Assimakopoulos (2022), where disaggregated forecasting was substantially harder than aggregate forecasting across all submitted methods. A full hierarchy-level breakdown for all four models is noted as a reproducibility improvement for future work.

4.8 Feature-Ablation Results

4.8.1 Feature Rankings by Demand Type

The ablation experiment removed one complete feature group at a time. A positive Δ WRMSSE means removing the group increased error (the group was beneficial); a negative value means removal reduced error.

Table 4.11: Mean Δ WRMSSE by feature group and demand type

Feature group	Smooth	Erratic	Intermittent	Lumpy
Lag	+0.0268	+0.0182	-0.0736	-0.0079
Rolling	-0.0011	+0.0079	-0.0504	-0.0474
Calendar	+0.1259	+0.1132	+0.0071	+0.0450
Event/SNAP	+0.0061	-0.0040	-0.0500	+0.0759
Price	-0.0079	+0.0042	+0.0267	+0.0950

Calendar was the most beneficial group for smooth (+0.1259) and erratic (+0.1132) demand. Price ranked first for lumpy demand (+0.0950) and intermittent demand (+0.0267). Three groups actively hurt XGBoost when present. Removing lag features improved intermittent WRMSSE by 0.0736 in all five folds, and removing lag also slightly improved lumpy WRMSSE (-0.0079). Removing rolling features improved both intermittent (-0.0504) and lumpy (-0.0474) WRMSSE. Removing event/SNAP improved intermittent WRMSSE by 0.0500 in four of five folds. Rolling removal produced a negligible improvement to smooth WRMSSE (Δ WRMSSE = -0.0011), indicating near-zero conditional contribution for smooth demand, a negligible effect.

4.9 SHAP Model-Reliance Results

Exact native TreeSHAP values were calculated for all 112,000 final XGBoost forecasts. The SHAP forecasts matched the saved XGBoost forecasts exactly (maximum difference: zero).

Table 4.12: Normalised absolute SHAP shares by demand type (%)

Demand type	Lag	Rolling	Calendar	Event/SNAP	Price
Smooth	9.09	69.71	12.95	4.98	3.27
Erratic	8.62	66.27	15.94	5.47	3.72
Intermittent	5.92	70.84	13.81	2.36	7.07
Lumpy	6.76	72.19	11.39	4.16	5.50

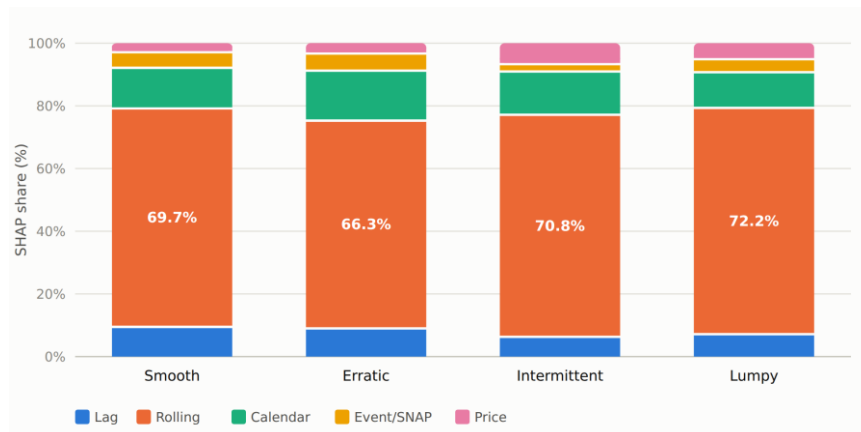


Figure 4.9: Normalised absolute SHAP shares by feature group and demand type, averaged across five folds and 200 series per demand type. Rolling features dominate model reliance across all four demand types.

Rolling features dominated model reliance in every demand type, ranging from 66.27% (erratic) to 72.19% (lumpy). Rolling ranked first in all 20 final XGBoost models. Calendar ranked second for all four demand types.

4.10 Comparison of SHAP and Ablation Evidence

The SHAP and ablation results agreed on some feature groups and diverged on others. Rolling features dominated SHAP reliance (66–72% of total attribution) but did not rank first in the ablation analysis for any demand type. For lumpy demand, rolling removal improved WRMSSE by 0.0474 despite accounting for 72.19% of SHAP reliance—the clearest example of high model reliance without forecasting benefit.

Calendar information showed the strongest agreement between the two methods: it ranked second by SHAP share and first in the smooth and erratic ablation analyses. Price showed the opposite of rolling: it received low SHAP shares (3.27–7.07%) but ranked first in the intermittent and lumpy ablation results. Event/SNAP showed low SHAP reliance across all demand types but opposite forecasting effects: removal improved intermittent WRMSSE by 0.0500 while worsening lumpy WRMSSE by 0.0759 in every fold.

The comparison confirms that SHAP reliance and forecast benefit are not interchangeable. Using both methods creates a more defensible interpretation of feature behaviour than reporting a single importance ranking.

4.11 LSTM Feasibility Results

The LSTM used a 56-day input sequence of sales history only and produced a direct 28-day forecast. It did not receive calendar, event, SNAP, price or engineered lag/rolling features.

Table 4.13: LSTM feasibility results (mean WRMSSE)

Demand type	Seasonal naive	XGBoost	LSTM	LSTM vs XGBoost	LSTM vs naive
Smooth	0.8064	0.6360	0.9278	45.88% worse	15.06% worse
Erratic	0.9500	0.7938	1.2513	57.64% worse	31.72% worse

The LSTM did not produce competitive WRMSSE. It lost to both XGBoost and seasonal naive in all ten fold comparisons. It did reduce MAE and RMSE relative to seasonal naive, but those bottom-level improvements did not translate into stronger hierarchy-weighted performance. The LSTM also showed systematic underforecasting: mean forecast error was -1.096 for smooth demand and -1.681 for erratic demand. This result applies to the tested sequence-only architecture, selected demand types and available computational budget; it should not be generalised to all deep-learning approaches.

4.12 Integrated Findings by Research Question

RQ1: Model Performance Across Demand Types

XGBoost reduced mean WRMSSE for every demand type relative to the seasonal naive benchmark: smooth 21.13%, erratic 16.44%, lumpy 10.44%, intermittent 0.50%. However, the four-model comparison revealed that Croston and SBA outperformed XGBoost for intermittent and lumpy demand. The best model therefore depended on demand structure: XGBoost for smooth and erratic demand, SBA or Croston for intermittent and lumpy demand.

RQ2: Feature-Group Contribution

No universal feature ranking was identified. Calendar was the strongest contributor for smooth ($+0.1259$) and erratic ($+0.1132$) demand. Price ranked first for lumpy ($+0.0950$) and intermittent ($+0.0267$) demand. Three groups hurt performance when present: removing lag features improved intermittent WRMSSE by 0.0736 in all five folds and slightly improved lumpy WRMSSE (-0.0079); removing rolling features improved both intermittent (-0.0504) and lumpy (-0.0474) WRMSSE; and removing event/SNAP features improved intermittent WRMSSE by 0.0500 and erratic WRMSSE by 0.0040.

RQ3: SHAP Reliance and Ablation

Rolling features dominated SHAP reliance (66–72%) but were not always beneficial. Price and event features sometimes received small SHAP attributions while providing strong protection against out-of-sample error. The comparison confirms that model reliance and out-of-sample contribution measure different properties.

RQ4: LSTM Feasibility

The restricted sequence-only LSTM (56-day input, no calendar/event/SNAP/price features) was not competitive according to WRMSSE. It was 45.88% worse than XGBoost for smooth demand and 57.64% worse for erratic demand.

RQ5: Practical Model-Selection Guidance

The combined evidence indicates that model selection should be demand-type-specific rather than based on a single forecasting method for all retail series. Model selection, feature design and interpretation method should reflect demand structure. SHAP should not be used alone to decide which features to retain.

XGBoost is the preferred model for smooth and erratic demand based on lowest observed WRMSSE across all five folds.

Croston and SBA are competitive alternatives for intermittent and lumpy demand where stability and simplicity matter.

SHAP importance should be treated as model reliance, not proof of forecast benefit.

Feature groups should be validated through ablation before being retained in production models.

LSTM findings should be treated as feasibility evidence only, not a general conclusion about deep learning.

4.13 Forecast Error by Horizon Day

Figure 4.10 plots mean RMSE by forecast horizon day for each demand type, averaged across 200 series and five folds. The error pattern across the 28-day horizon is not a monotonic increase — instead all four demand types show a cyclical pattern with spikes at approximately days 6–7, 13–14, 20–21 and 27–28. This weekly periodicity suggests that XGBoost error is driven primarily by day-of-week structure rather than by recursive error accumulation. If recursive propagation were the dominant mechanism, error would increase steadily across the horizon; the observed pattern instead reflects that certain forecast days — consistently weekend-adjacent positions — are harder to predict regardless of horizon distance. Intermittent demand shows the sharpest within-week variation, with RMSE rising from 1.04 at day 1 to 3.65 at day 7 before partially recovering. This is consistent with the sparse-demand mechanism discussed in Section 5.3 — intermittent series have few non-zero observations, and the model struggles most at horizon positions that fall on high-activity days. The horizon-day analysis does not support the recursive error propagation explanation as the primary driver of XGBoost's weak intermittent performance; it instead points to structural demand periodicity as the more likely source.

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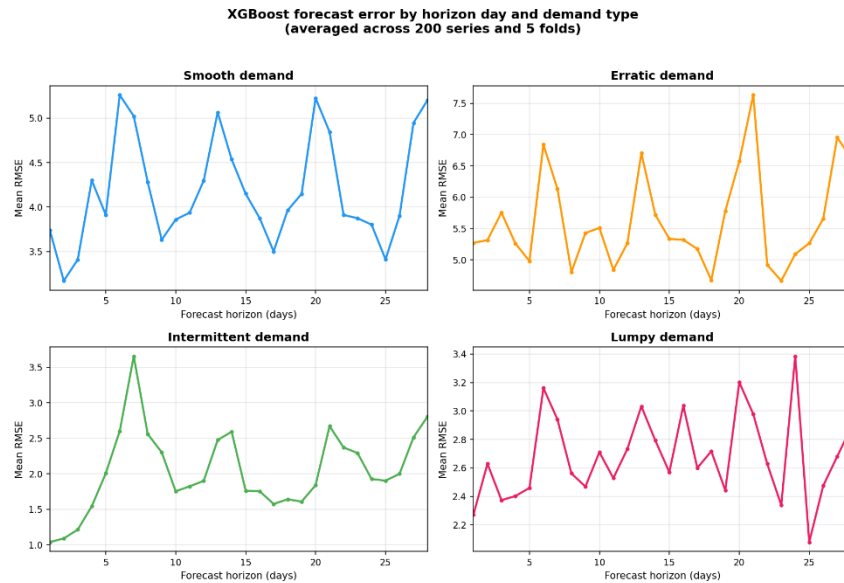


Figure 4.10: XGBoost forecast error by horizon day and demand type, averaged across 200 series and five folds. RMSE is shown on the y-axis; lower values indicate better forecast accuracy. The cyclical pattern with weekly spikes is consistent across all four demand types.

4.14 Chapter Summary

The full M5 population was dominated by sparse demand (95.156% intermittent or lumpy). XGBoost improved on seasonal naive for all four demand types but by very different amounts. The four-model comparison showed that Croston and SBA outperformed XGBoost for intermittent and lumpy demand, while XGBoost was clearly stronger for smooth and erratic demand. Feature contribution and SHAP reliance varied across demand types, and the two measures produced different rankings. The restricted LSTM was not competitive under its constrained design. Forecasting performance, model selection and feature usefulness all depend on demand structure.

Chapter 5: Discussion

5.1 Chapter Overview

This chapter interprets the findings presented in Chapter 4 in relation to the research questions and the literature reviewed in Chapter 2. The discussion addresses whether forecasting performance differed across the four demand types, which feature groups improved XGBoost accuracy, how SHAP model reliance compared with ablation evidence, and whether the restricted LSTM provided a useful alternative. It also examines the Croston/SBA results and the surprising finding that these methods helped for erratic demand.

5.2 Demand Heterogeneity and Forecasting Difficulty

The classification results confirmed that the lowest level of the M5 dataset is dominated by sparse demand. Intermittent and lumpy series represented 95.156% of all 30,490 item-store combinations, with an overall zero-sales rate of approximately 68.0%. This supports the argument of Fildes, Ma and Kolassa (2022) that retail demand forecasting should not be treated as one homogeneous problem.

The four demand groups were separated using both demand occurrence and positive-demand variability. Smooth and erratic series had similar mean ADI values (1.176 and 1.207) but different mean CV^2 values (0.376 versus 0.728). Intermittent demand had the highest zero-sales rate (72.436%) but comparatively stable positive quantities ($CV^2 = 0.286$). Lumpy demand combined both difficulties: a mean ADI of 4.165 with a high CV^2 of 0.721. These findings are consistent with the conceptual distinction established by Syntetos, Boylan and Croston (2005), applied here to a daily retail dataset two orders of magnitude larger than the original 3,000-series automotive study.

The seasonal naive benchmark demonstrated the effect of demand structure on forecasting difficulty. Smooth demand produced the lowest benchmark WRMSSE (0.8064), while lumpy demand produced the highest (1.2589). The intermittent result—low MAE (0.8940) but high WRMSSE (1.2156)—illustrates why sparse demand can appear easier under an unscaled absolute-error measure, supporting the argument of Hyndman and Koehler (2006) that scaled errors provide a fairer basis for comparison.

5.3 XGBoost Performance Across Demand Types

XGBoost improved mean WRMSSE relative to seasonal naive for all four demand types, but the size of the improvement differed substantially: 21.13% for smooth, 16.44% for erratic, 10.44% for lumpy and 0.50% for intermittent demand. The smooth-demand result was the clearest evidence in favour of the feature-based design: the model won all five folds and reduced WRMSSE from 0.8064 to 0.6360. The strong result is broadly consistent with the M5 competition evidence (Makridakis, Spiliotis and Assimakopoulos, 2022), though the present study adds the detail that improvement depended on demand pattern even within one item-store sample.

Erratic demand also benefited consistently (5/5 fold wins), suggesting that high positive-demand variability does not prevent machine learning from adding value when observations remain frequent. The intermittent result was the weakest: the mean improvement was only 0.50%, and long zero-sales intervals provided limited information for recursive lag and rolling features.

The fold-level results revealed two distinct patterns. Fold 4 was XGBoost's strongest fold across all four demand types simultaneously (smooth 0.5389, erratic 0.6448, intermittent 0.9247, lumpy 0.8492), each the lowest WRMSSE in any fold for that demand type. By contrast, Fold 1 was where both of XGBoost's WRMSSE losses

against the seasonal naive benchmark occurred (intermittent and lumpy). These are different folds, different comparisons and different directions of deviation—one instance of anomalous strength, one of anomalous weakness. With only five forecast origins, it is not possible to determine whether these reflect genuine temporal variation in XGBoost’s relative advantage or sampling noise.

5.4 Croston, SBA and the Frequent-Demand Surprise

The most unexpected finding from the four-model comparison was the performance of Croston and SBA for erratic demand. Erratic series have ADI below 1.32—they sell frequently—and were not the original target population for methods designed for intermittent demand. Yet both methods improved on the seasonal naive benchmark: Croston reduced mean WRMSSE from 0.9500 to 0.8819 and SBA from 0.9500 to 0.8853.

The mechanism behind this result appears to involve demand-size variability rather than demand sparsity. Erratic series sell on most days but with highly variable quantities (mean $CV^2 = 0.728$). The seasonal naive benchmark repeats the final seven training days exactly, including any unusually high or low observations from that particular week. When the final week contained an atypical demand spike or dip, the naive forecast propagated that anomaly across the entire 28-day horizon. Croston and SBA, by contrast, apply exponential smoothing to the non-zero demand sizes, producing a more stable flat-rate estimate that dampens the influence of recent outliers.

This finding reframes the role of Croston-type methods. The conventional interpretation is that they help because they handle long zero intervals (high ADI). The erratic-demand result shows they can also help by stabilising demand-level estimation when positive quantities are highly variable (high CV^2). The underlying mechanism—exponential smoothing of demand size—addresses variability regardless of whether the variability is accompanied by sparsity. Croston and SBA therefore help wherever demand deviates from smooth regularity, whether via sparsity, via demand-size variability, or via both.

For intermittent demand, Croston (1.1156) and SBA (1.0978) both outperformed XGBoost (1.2095) and the seasonal naive benchmark (1.2156). SBA recorded the lowest WRMSSE of any model for intermittent demand. For lumpy demand, Croston (1.0674) was the best-performing model, followed by SBA (1.0756), both outperforming XGBoost (1.1275). The only demand type where Croston and SBA performed worse than the seasonal naive benchmark was smooth demand, where the flat-rate forecast could not represent the regular weekly structure that the seasonal naive model captured directly. Beyond the average accuracy advantage established in Chapter 4, Croston and SBA also show consistently lower fold-to-fold variability than XGBoost for intermittent and lumpy demand (Table 4.10), meaning these methods are not only more accurate on average for sparse demand but also more reliably so across different time periods.

The mechanism proposed here — that exponential smoothing of demand size reduces the influence of recent outlier observations on the flat-rate forecast — is consistent with the ablation and fold-level evidence but has not been directly tested at the series level. A direct test would examine whether Croston and SBA gains for erratic demand are concentrated in series with the highest positive-demand CV^2 , the most recent anomalous demand spikes, or the greatest instability in the final training week used by the seasonal naive benchmark. The present study reports group-level means across 200 erratic series and cannot establish whether the proposed mechanism operates continuously across the CV^2 dimension or only above a particular variability threshold.

This is noted as a hypothesis supported by the group-level evidence rather than a verified causal explanation.

5.5 Interpretation of the Feature-Ablation Results

The ablation results showed that the usefulness of retail information varied across demand types. Calendar ranked first for smooth (+0.1259) and erratic (+0.1132) demand, consistent with the emphasis on calendar and seasonal effects in the retail forecasting literature (Fildes, Ma and Kolassa, 2022). Price ranked first for lumpy (+0.0950) and intermittent (+0.0267) demand, despite higher missing-price rates for these groups (20.742% intermittent, 18.890% lumpy).

The intermittent lag result was the clearest single finding: removal improved WRMSSE by 0.0736 in all five folds. Intermittent lag features were mostly zeros, and the XGBoost model learned from noise rather than genuine demand signal. Recursive forecasting likely amplified the problem as each day's prediction fed into the next day's lag features. The same pattern appeared for lumpy demand, where removing rolling features improved WRMSSE by 0.0474 and removing lag produced a small improvement of 0.0079. Intermittent rolling removal also improved WRMSSE (−0.0504), confirming that recent-history features were counterproductive for sparse demand. A possible explanation is that correlated lag and rolling features introduce redundant information that the model overfits to during training; removing one group may reduce this redundancy and improve generalisation to unseen test periods.

Event/SNAP information produced opposite effects across groups: removal improved erratic WRMSSE slightly (by 0.0040) and improved intermittent WRMSSE by 0.0500, while worsening lumpy WRMSSE by 0.0759 in every fold. Event/SNAP removal worsened WRMSSE by 0.0061 for smooth demand, indicating a small but consistent contribution from event and SNAP features for that demand type. The same low level of average model reliance therefore corresponded to opposite forecasting effects depending on demand structure.

The SHAP and ablation evidence together suggest a plausible explanation for why Croston and SBA outperformed XGBoost specifically on intermittent demand. XGBoost's SHAP results show rolling and lag features accounting for the majority of model reliance for intermittent demand, despite intermittent series being 72.4% zero-sales by observation. The ablation shows that removing lag improved intermittent WRMSSE by 0.0736 and removing rolling improved it by 0.0504—the features XGBoost relied on most were actively hurting its forecasts. The likely mechanism: with demand this sparse, lag and rolling windows mostly encode strings of zeros, so XGBoost learns from noise rather than genuine demand signal. Croston and SBA avoid this structurally. They do not use lag or rolling windows at all; instead, they separate demand size from demand timing explicitly rather than inferring that separation from a noisy recent-history window. This is a plausible explanation for why Croston and SBA beat XGBoost specifically where XGBoost's own feature reliance is most disconnected from what actually helps. Croston and SBA separate demand occurrence from demand size, which means their forecasts are not distorted by strings of zeros in the same way that lag and rolling features in XGBoost are — the structural simplicity of the method becomes an advantage when the signal in recent history is weak.

Kumar et al. (2020) show that conditional Shapley methods, including TreeSHAP, estimate a feature's importance from the model's behaviour across training examples where that feature happens to take a particular value, not from directly testing what happens if only that feature is changed. When two features are correlated, the one that is easier for the tree structure to split on can absorb credit that rightfully belongs to the

feature actually driving the outcome. This offers a plausible explanation for the SHAP-ablation divergence observed for intermittent demand in this study. Rolling and lag features do not directly cause demand to occur; they mark whether a product is currently in an active selling phase rather than effectively dead. For a series with a 72.4% zero-sales rate, this distinction between ‘active but currently zero’ and ‘no longer selling’ carries real predictive value, and XGBoost’s tree splits can exploit it. But because this signal is correlated with, rather than causally responsible for, the timing and size of the next demand event, TreeSHAP appears to award rolling and lag features credit for information the model is using indirectly rather than mechanistically. This is consistent with Kumar et al.’s (2020) broader argument that conditional Shapley values can attribute importance to features with no direct interventional effect on a model’s output, and it supports treating SHAP results in this study as a measure of model reliance rather than a causal or unconditional statement of feature value.

A direct test of whether recursive error propagation explains XGBoost’s weakness on sparse demand would require reporting forecast errors by horizon day — if XGBoost error grows more sharply across the 28-day window for intermittent and lumpy demand than for smooth and erratic demand, that pattern would support the propagation mechanism. Specifically, the ablation result that removing lag features improved intermittent WRMSSE by 0.0736 in all five folds is consistent with the propagation argument — lag features updated with erroneous zero forecasts on early horizon days would corrupt the rolling and lag inputs for later days — but this consistency does not constitute a direct test. Future work examining error growth by horizon day and by time-since-last-nonzero-demand would provide stronger evidence for or against the recursive propagation explanation.

5.6 SHAP Reliance and Forecast Benefit

Rolling features dominated the internal behaviour of every final XGBoost model (66–72% of total absolute SHAP attribution) and ranked first in all 20 demand-type-fold models. However, rolling features did not rank first in the ablation results for any demand type. For lumpy demand, removing them improved WRMSSE despite accounting for 72.19% of the SHAP share—the clearest example of the distinction between fitted-model reliance and predictive value.

Price showed the opposite pattern: low SHAP shares (3.27–7.07%) but ranked first in the intermittent and lumpy ablation analyses. Calendar showed the strongest agreement between the two methods, with both strong SHAP reliance and clear forecast benefit for smooth and erratic demand. This combination of agreements and disagreements shows why model explanation and predictive usefulness should be analysed separately (Lundberg and Lee, 2017; Lundberg et al., 2020).

High SHAP importance means the model relied on a feature during prediction; it does not mean the feature improved out-of-sample forecast accuracy. The ablation results provide the appropriate evidence for forecast benefit.

5.7 Hierarchy and Metric Effects

The hierarchy-level results demonstrated that model comparison depends on where the forecast is evaluated. XGBoost reduced MAE and RMSE in all 20 demand-type-fold comparisons, but lost two WRMSSE comparisons. For intermittent demand, XGBoost was weaker across Levels 1 to 6 but stronger across Levels 7 to 12. These findings are consistent with the M5 competition evidence that performance changes across hierarchy levels (Makridakis, Spiliotis and Assimakopoulos, 2022) and support Tashman’s (2000) argument that forecasting methods should be tested across multiple origins and clearly defined groups.

5.8 Discussion of the LSTM Feasibility Experiment

The sequence-only LSTM did not provide competitive WRMSSE for either smooth or erratic demand. It was 45.88% worse than XGBoost and 15.06% worse than seasonal naive for smooth demand; 57.64% worse than XGBoost and 31.72% worse than seasonal naive for erratic demand. It showed systematic underforecasting (mean errors of -1.096 and -1.681) in all ten fold models.

The comparison with XGBoost was not a controlled architecture test: XGBoost received 18 engineered features including calendar, event, SNAP and price information, while the LSTM received only the previous 56 daily sales observations. The LSTM also did not implement the probabilistic likelihoods, covariates, scaling and extensive tuning used in more advanced systems such as DeepAR (Salinas et al., 2020). The correct conclusion is therefore narrow: the selected sequence-only LSTM was not competitive under the completed experimental conditions. The result does not establish that LSTM or neural forecasting generally is unsuitable for retail demand.

5.9 Research Contribution and Interpretation Boundaries

The main contribution of this dissertation is not a new forecasting algorithm but the combination of established methods within one demand-type-specific experimental design. The study connected ADI/CV² classification with large-scale daily retail forecasting, evaluated four models separately for each demand type, tested five feature groups through ablation, compared exact TreeSHAP reliance with ablation evidence, and conducted a transparent LSTM feasibility study.

The contributions remain subject to five boundaries. The balanced sample did not reproduce the natural M5 population. Demand categories were fixed retrospectively. The hierarchy was reconstructed within the subset. Only one XGBoost configuration and one LSTM architecture were evaluated. The five forecast origins were insufficient for formal significance claims. These boundaries define what the evidence can support: a controlled account of how performance and feature behaviour varied within the selected sample, not a universal claim about the best retail forecasting model.

As a limited sensitivity check, XGBoost bottom-level forecast errors were re-examined for the 750 series whose full-history and earliest-origin demand labels agreed. XGBoost per-series forecast errors were available for all 800 series across all five folds. Restricting the comparison to the 750 stable-label series — 189 smooth, 179 erratic, 197 intermittent and 185 lumpy — produced fold-level RMSE patterns consistent with the full-sample results in every demand type and every fold. No fold changed direction relative to the full-sample result. Because per-series Croston and SBA forecast values were not saved at the series level, a full WRMSSE reconstruction for the stable-label subset could not be performed without model retraining; the XGBoost stability check therefore provides partial but consistent evidence that the headline rankings are not driven by the 50 series whose classification changed between full-history and training-only labelling. This result is reported as a sensitivity check rather than a replacement for the primary analysis.

5.10 Chapter Summary

The demand classification confirmed that sparse demand dominated the M5 item-store population. XGBoost improved mean WRMSSE for every demand type but by very different amounts. The four-model comparison revealed that Croston and SBA outperformed XGBoost for intermittent and lumpy demand, and—unexpectedly—also improved on the seasonal naive benchmark for erratic demand, where their smoothed demand-level estimation reduced the effect of high positive-demand variability.

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Feature contribution and SHAP reliance varied across demand types, and the two measures produced different rankings. The restricted LSTM was not competitive under its constrained design. The overall conclusion is that model performance, feature usefulness and model reliance all depend on demand structure.

Chapter 6: Conclusion and Recommendations

6.1 Chapter Overview

This chapter concludes the dissertation by summarising the completed research, answering the five research questions and presenting the main academic and practical contributions. It also provides recommendations for retail forecasting practice, identifies the main limitations and outlines areas for future research.

6.2 Summary of the Research

The dissertation examined whether forecasting performance and feature behaviour differed across smooth, erratic, intermittent and lumpy demand within the Walmart M5 dataset. All 30,490 item-store series were classified using ADI and CV^2 , producing 983 smooth, 494 erratic, 23,075 intermittent and 5,938 lumpy series. Intermittent and lumpy demand together represented 95.156% of the full population, and 68.0% of all item-store-day observations recorded zero sales.

A balanced experimental subset of 200 series per demand type (800 total) was used for the forecasting experiments. Four forecasting models were compared: the seasonal naive benchmark, Croston's method, the Syntetos–Boylan Approximation and demand-type-specific XGBoost. All models were evaluated using five expanding walk-forward folds with a fixed 28-day test window. The feature-ablation stage produced 100 ablation models and 560,000 forecasts. Exact TreeSHAP values were calculated for all 112,000 final XGBoost forecasts. A restricted sequence-only LSTM feasibility experiment was conducted for smooth and erratic demand.

6.3 Conclusions by Research Question

6.3.1 Research Question 1

RQ1: How does the forecasting performance of the seasonal naive baseline, Croston's method, SBA, and XGBoost differ across smooth, erratic, intermittent, and lumpy demand types?

Table 6.1: Four-model WRMSSE comparison by demand type

Demand type	Seasonal naive	Croston	SBA	XGBoost	Best model
Smooth	0.8064	0.8556	0.8536	0.6360	XGBoost
Erratic	0.9500	0.8819	0.8853	0.7938	XGBoost
Intermittent	1.2156	1.1156	1.0978	1.2095	SBA
Lumpy	1.2589	1.0674	1.0756	1.1275	Croston

Model suitability depended on demand structure. XGBoost was the strongest model for smooth and erratic demand, where frequent observations provided a continuous signal for its 18-feature recursive design. For intermittent and lumpy demand, Croston and SBA outperformed both XGBoost and the seasonal naive benchmark. Croston and SBA also improved on the seasonal naive benchmark for erratic demand, despite erratic series having ADI below 1.32. This suggests that these methods help wherever demand deviates from smooth regularity—via sparsity, via demand-size variability, or via both—not only for the sparse-demand categories where they were originally designed.

6.3.2 Research Question 2

RQ2: Which feature groups—lag, rolling, calendar, event/SNAP, and price—contribute most to XGBoost forecasting performance for each demand type, as measured by a within-type ablation study?

Feature contribution was demand-type-specific. Calendar provided the largest ablation benefit for smooth (+0.1259) and erratic (+0.1132) demand. Price ranked first for lumpy (+0.0950) and intermittent (+0.0267) demand. Three groups actively hurt performance: lag removal improved intermittent WRMSSE in all five folds (−0.0736) and slightly improved lumpy WRMSSE (−0.0079). Rolling removal improved both intermittent (−0.0504) and lumpy (−0.0474) WRMSSE. Event/SNAP removal improved intermittent WRMSSE in four of five folds (−0.0500).

6.3.3 Research Question 3

RQ3: How do the main SHAP feature drivers differ across smooth, erratic, intermittent, and lumpy demand, and how does SHAP model reliance compare with the ablation evidence?

Rolling features dominated SHAP reliance across every demand type (66.27–72.19% of total absolute attribution) and ranked first in all 20 final XGBoost models. However, SHAP reliance and ablation evidence did not always agree. Lumpy models assigned 72.19% of reliance to rolling features even though removing them improved WRMSSE. Price received low SHAP shares but ranked first in the intermittent and lumpy ablation results. Calendar produced the clearest agreement: second-highest SHAP share and strongest ablation benefit for smooth and erratic demand.

6.3.4 Research Question 4

RQ4: Does a limited LSTM feasibility study add forecasting value over XGBoost for smooth and erratic demand types?

The restricted sequence-only LSTM (56-day input, no calendar/event/SNAP/price features) was not competitive. For smooth demand, it was 45.88% worse than XGBoost and 15.06% worse than seasonal naive. For erratic demand, it was 57.64% worse than XGBoost and 31.72% worse than seasonal naive. It showed systematic underforecasting in all ten fold models. The weak result applies to the specific architecture and information set tested; it should not be generalised to all neural forecasting methods.

6.3.5 Research Question 5

RQ5: How can the combined model-performance, feature-ablation and SHAP results inform practical model-selection guidance for retail demand planning?

The findings support a demand-type-specific approach. For smooth demand, the full XGBoost model provided the strongest result; calendar information was particularly important. For erratic demand, XGBoost also performed strongly, with Croston and SBA providing a simpler alternative that outperformed the seasonal naive benchmark. For intermittent demand, Croston and SBA outperformed XGBoost; if XGBoost is used, removing lag (−0.0736), rolling (−0.0504) and event/SNAP (−0.0500) all improved WRMSSE, so a reduced feature set is preferable. For lumpy demand, Croston provided the best result, while price (+0.0950) and event/SNAP (+0.0759) were the most beneficial XGBoost features; lag and rolling both slightly hurt performance. SHAP should be used to explain model behaviour but not as the sole basis for feature retention; feature-removal testing remains necessary.

6.4 Main Contributions

The study makes five connected contributions. First, it connected ADI/CV² demand classification with large-scale daily retail forecasting by classifying all 30,490 M5 item-store series before model comparison. Second, four forecasting models were evaluated separately for each demand type, revealing that Croston and SBA outperformed XGBoost for intermittent and lumpy demand while XGBoost was strongest for smooth and erratic demand. Third, the demand-type-specific ablation showed that feature value was conditional rather than universal. Fourth, the direct comparison of exact TreeSHAP and ablation demonstrated that model reliance and forecast benefit can produce different conclusions. Fifth, the transparent reporting of the negative LSTM result shows that model complexity alone does not guarantee improved forecasting.

6.5 Practical Recommendations

Retailers should classify demand using ADI and CV² before comparing models. For frequent-demand products, calendar information should be prioritised in feature-based models. For intermittent products, Croston-type methods should be considered as alternatives to complex machine-learning models; lag, rolling and event/SNAP features should be tested carefully because all three hurt intermittent WRMSSE in the ablation. For lumpy demand, price (+0.0950) and event/SNAP (+0.0759) were the most beneficial XGBoost features, while lag and rolling features both slightly hurt performance; Croston-type methods provide strong baselines. Retailers should evaluate multiple metrics (WRMSSE, MAE, RMSE) and hierarchy levels, combine SHAP with performance testing, monitor forecast bias, and match model complexity to practical benefit.

6.6 Limitations

Because the evaluation contained only five forecast origins and fold observations were not independent, formal significance testing was not used. Differences are therefore interpreted descriptively through error magnitude, fold consistency and demand-type patterns.

The analysis used one retailer's data from three US states. The balanced sample of 800 series did not represent the natural M5 distribution. Demand categories were calculated retrospectively using the complete sales history. Zero sales could not be separated from product unavailability. Price information was incomplete for intermittent and lumpy demand. Only one XGBoost configuration and one LSTM architecture were evaluated. Feature-ablation models were not retuned after feature removal. The hierarchy was reconstructed within the balanced subset. Only five forecast origins were evaluated, which was insufficient for formal statistical inference. The study evaluated forecast accuracy rather than inventory cost, waste, service level or profit.

The Croston and SBA smoothing parameter was fixed at $\alpha = 0.10$ by convention rather than optimised through cross-validation. Sensitivity testing across 0.05 to 0.20 showed that for SBA on intermittent demand, $\alpha = 0.15$ and $\alpha = 0.20$ both produced lower WRMSSE than $\alpha = 0.10$. The headline finding held across the full tested range, but the reported SBA intermittent result is conservative—tuning could have produced a stronger outcome.

The demand classification used throughout this study was determined retrospectively using the complete 1,941-day sales history for each series. In a deployed forecasting system, demand type would be assigned using only history available at the forecast origin, and class membership may shift as new observations accumulate. A

classification stability audit conducted on all 800 selected series found 93.8% agreement between full-history and training-only classifications, with intermittent demand the most stable (98.5%) and erratic demand the least stable (89.5%). Despite this broad stability, 415 series fell within 0.10 of at least one classification threshold, and results for erratic demand in particular should be interpreted with the awareness that a portion of those series may have carried a different label under a stricter origin-safe design. The study's strongest conclusions — that Croston and SBA outperform XGBoost for intermittent and lumpy demand — rest on the two most stable demand classes and are therefore less sensitive to this limitation than the erratic-demand findings.

The balanced experimental sample of 200 series per demand type was constructed using a single random seed (42) and sampling fractions that ranged from 0.87% for intermittent to 40.49% for erratic demand. The category and state composition of the balanced sample was not formally verified against the full M5 population within each demand type, meaning that the selected series may not be representative of the broader demand-type population even within the M5 dataset. Results should therefore be treated as evidence about the specific series evaluated rather than as population estimates for each demand type across the full Walmart item-store catalogue. Stability across independently drawn cohorts was not assessed in this study and represents a priority for future work.

A within-sample cohort stability check was conducted to assess whether XGBoost performance conclusions were sensitive to the specific 200 series selected per demand type. Two additional subsamples of 180 series per demand type were drawn from the original 800 selected series using pre-specified seeds of 123 and 456, and XGBoost forecast errors were recomputed for each subsample. Bottom-level RMSE varied by at most 0.23 units for smooth demand, 0.16 for erratic, 0.09 for intermittent and 0.03 for lumpy demand across the three cohorts. The rank ordering of XGBoost performance across demand types was identical in all three cohorts: smooth demand consistently produced the strongest relative improvement and intermittent demand the weakest. This check provides limited but consistent evidence that the XGBoost performance pattern is not an artefact of the specific series selected, while acknowledging that a full stability assessment would require rerunning all four models on independently drawn cohorts from the complete 30,490-series M5 population.

6.7 Recommendations for Future Research

Future work could evaluate the full 30,490-series M5 population, use dynamic demand classification recalculated at each forecast origin, compare additional methods (exponential smoothing, LightGBM, direct multi-horizon models, hurdle or two-stage occurrence-and-size models), test alternative multi-step strategies, retune ablated models independently, examine feature interactions, develop stronger neural benchmarks with covariates and probabilistic likelihoods, increase the number of forecast origins, include inventory and cost outcomes, and investigate missing price and stock availability information.

6.8 Personal Reflection

Early in the project I found a bug in my seasonal naive implementation. The baseline was using test-period observations after the first seven forecast days to update later forecasts, which meant days 8–28 were evaluated against a baseline that had already seen some of the answer. Smooth-demand WRMSSE looked implausibly low. I caught it by plotting the naive predictions day by day and noticing the suspiciously tight fit from day 8 onwards. After correcting the code so that the entire 28-day forecast was

generated from one fixed training-period origin, I re-ran every downstream comparison from scratch. The episode changed how I worked for the rest of the project: every new model was checked against a deliberately wrong input first (a shuffled test set, a zeroed feature) to confirm that the evaluation pipeline responded as expected. Without that bug I might have trusted my pipeline too easily and missed subtler problems later.

The LSTM result was the other formative experience. I expected the sequence-only LSTM to be at least competitive with XGBoost on smooth demand, where sales are frequent and weekly patterns are strong. It was 45.88% worse. My first reaction was to assume a coding error—perhaps the input scaling was wrong, or the direct 28-day output layer was collapsing to a constant. I checked both: the inputs were correctly scaled per series, and the output layer produced varying predictions across forecast days and series. The model was simply not learning the weekly structure from raw sales alone, and its systematic underforecasting (mean error -1.096 for smooth demand) showed that the 56-day window was too short or too noisy to recover the signal that calendar features gave XGBoost for free. Accepting this as a genuine negative result rather than inflating it with caveats about “future potential” was uncomfortable but necessary. It taught me that honest reporting of what did not work is more useful than hedging a weak result until it sounds like a partial success.

6.9 Final Conclusion

This dissertation investigated retail demand forecasting under demand heterogeneity using the Walmart M5 dataset. The central finding is that forecasting performance, model suitability and feature usefulness depend strongly on the structure of demand. XGBoost was most effective for smooth and erratic demand, while Croston and SBA outperformed it for intermittent and lumpy demand. Croston and SBA also helped for erratic demand, where their smoothed demand-level estimation reduced the effect of high positive-demand variability. Feature contribution varied across demand types, and SHAP model reliance did not equal forecast benefit. The restricted LSTM was not competitive under its constrained design. A demand-aware forecasting strategy is therefore more defensible than applying one model and one feature set uniformly across every retail series.

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Appendices

Appendix A: Croston and SBA Fold-Level Results

Table A.1: Croston fold-level WRMSSE by demand type

Fold	Smooth	Erratic	Intermittent	Lumpy
1	0.8970	0.9781	1.0045	1.0482
2	0.8977	0.9741	0.9628	1.1815
3	0.8146	0.7907	1.2437	1.0538
4	0.8106	0.8552	1.1090	1.0325
5	0.8581	0.8115	1.2580	1.0210
Mean	0.8556	0.8819	1.1156	1.0674

Table A.2: SBA fold-level WRMSSE by demand type

Fold	Smooth	Erratic	Intermittent	Lumpy
1	0.9004	1.0086	0.9540	1.0786
2	0.9380	1.0159	0.9382	1.2079
3	0.8148	0.7900	1.2716	1.0558
4	0.7616	0.8168	1.1273	1.0036
5	0.8534	0.7949	1.1978	1.0323
Mean	0.8536	0.8853	1.0978	1.0756

Appendix B: WRMSSE Denominator Audit

The denominator audit checked all WRMSSE scaling denominators across 90 model–demand-type–fold combinations. The maximum absolute difference between any pair of denominators for the same series and fold was 0.0, confirming identical scaling across all models.

Appendix C: Sampling Summary

Table C.1: Balanced subset sampling fractions

Demand type	Population	Selected	Sampling fraction
Smooth	983	200	20.35%
Erratic	494	200	40.49%
Intermittent	23,075	200	0.87%
Lumpy	5,938	200	3.37%